Policy Optimization: From Guarantees to LLMs

From Policy Gradients to Trust Regions & Preference Optimization

Yakov

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Part I — Main Sources

The Policy Gradient Theorem

Sutton et al. (1999)

The foundation that proved you can optimize a parameterized policy directly by estimating the gradient of the expected return.

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\right]$$



Neuro-Dynamic Programming

Bertsekas & Tsitsiklis (1996), Ch. 6

Formalized Approximate Policy Iteration (API). Chapter 6 is historically significant here because it exposed the fundamental instability of using function approximators (like neural networks) in reinforcement learning.

Conservative Policy Iteration

Kakade & Langford (2002)

Established the theoretical foundation for monotonic policy improvement using the performance difference lemma and mixture policies.

Trust Region Policy Optimization

Schulman et al. (2015)

Scaled CPI's concepts to deep neural networks by using a KL-divergence constraint to enforce a "trust region" during policy updates, preventing destructively large steps.

$$J(\pi_{new}) \geq L_{\pi_{old}}(\pi_{new}) - \frac{4\gamma\epsilon}{(1-\gamma)^2}\max_s D_{KL}(\pi_{old}(\cdot|s)\|\pi_{new}(\cdot|s))$$

Approximately Optimal Approximate Reinforcement Learning

Kakade & Langford
ICML 2002

Introduces **Conservative Policy Iteration (CPI)**

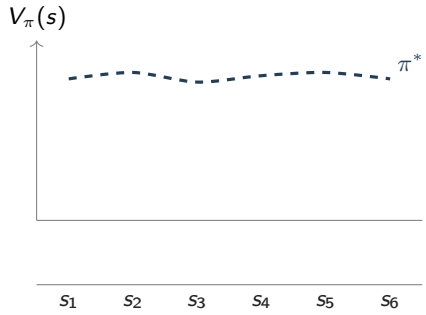
Setting & Assumptions

A finite MDP $(S, D, A, \mathcal{R}, P)$:

- **Finite** state and action spaces: $|S|, |A| < \infty$.
- **Bounded rewards**: $\mathcal{R} : S \times A \to [0, R]$.
- **Discounted**: $\gamma \in [0, 1)$.
- **Unknown dynamics**: $P(s'; s, a)$ is not given — we can only sample transitions (model-free).
- **Start-state distribution** D: the goal is

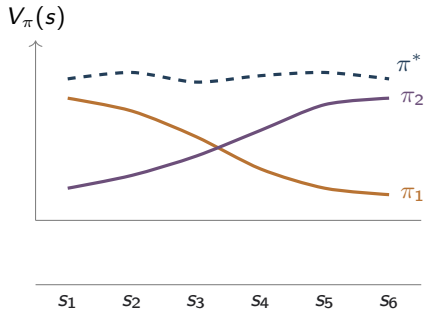
$$\eta_D(\pi) = \mathbb{E}_{s_0 \sim D}\left[V_\pi(s_0)\right]$$

The Start-State Distribution D



Exact: π^* on top everywhere
$\Rightarrow$ D irrelevant.

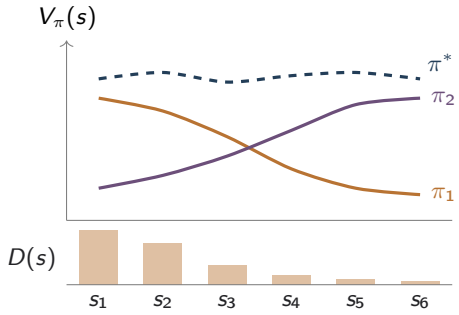
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Approximate: curves cross
restricted class $\{\pi_\theta\}$ · limited data

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Goal (Sec. 2): rank by D

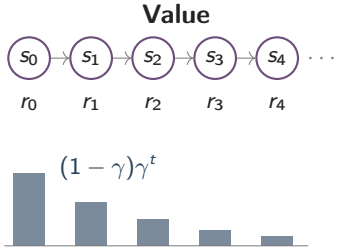
$$\eta_D(\pi) = \sum_s D(s)\, V_\pi(s)$$

Here π_1 wins.

D **comes from the application:** robot start poses, game openings, user prompts.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$



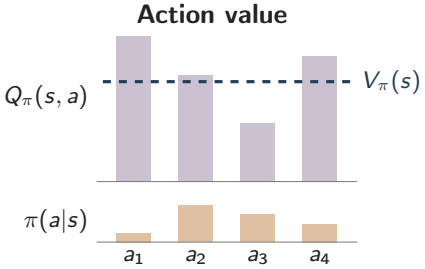
Weights sum to 1, so V is an average of rewards: $V \in [0, R]$.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$

$$Q_\pi(s,a) = (1-\gamma)\mathcal{R}(s,a) + \gamma\,\mathbb{E}_{s'}\big[V_\pi(s')\big]$$

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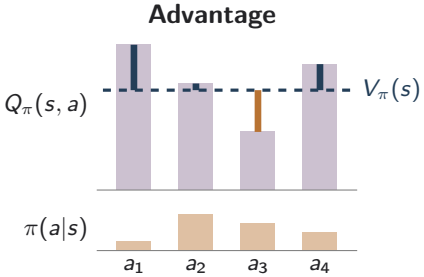
Take a first, then follow π.
V_π = bars averaged with weights $\pi(a|s)$.

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$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$
$$\mathbb{E}_{a\sim\pi}\big[A_\pi(s,a)\big] = 0, \quad \mathbb{E}_{a\sim\pi'}\big[A_\pi(s,a)\big] \neq 0$$



A = gap to the line. Weighted by π the gaps cancel.
A π' that favors the **positive** gaps gets $\mathbb{E}_{\pi'}[A_\pi] > 0$.

Notation

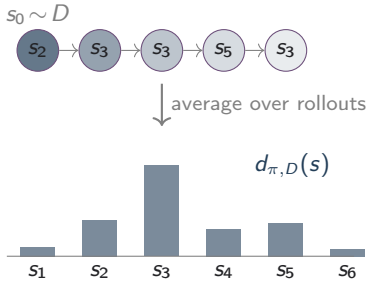
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$$d_{\pi,D}(s) = (1-\gamma)\sum_t \gamma^t \Pr(s_t = s;\, \pi, D)$$

Discounted visitation



Where π spends its time from $s_0 \sim D$, early steps weighted more.
Not the same as D (only s_0).

Notation

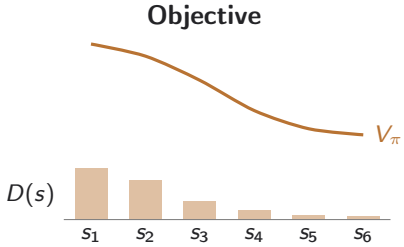
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$$d_{\pi,D}(s) = (1-\gamma)\textstyle\sum_t \gamma^t \,\mathrm{Pr}(s_t = s;\, \pi, D)$$

$$\eta_D(\pi) = \mathbb{E}_{s\sim D}\big[V_\pi(s)\big]$$



Average V_π over start states only.
One number to rank policies.

Three Questions Any Approximate Method Should Answer

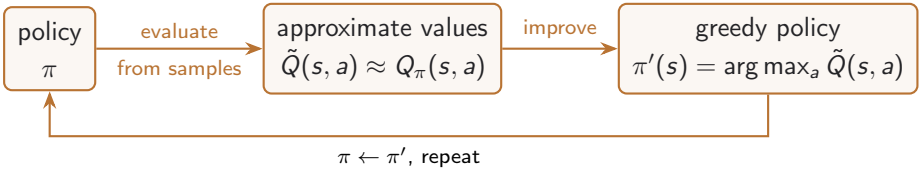
1. Is there a **performance measure** that is guaranteed to **improve at every step**?
2. How hard is it to **verify** that a given update improves this measure?
3. What **performance level** is reached after a **reasonable number** of updates?

Note: not "what is the asymptotic policy?" — only policies reachable in reasonable time.

Approximate value functions
Weak answers to all three.

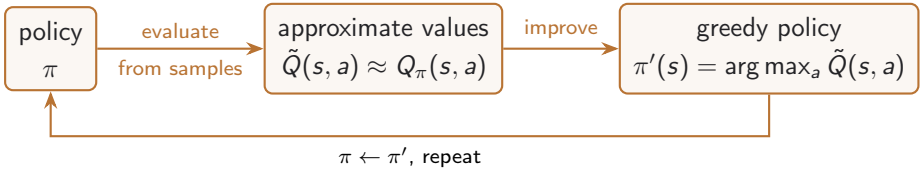
Policy gradients
Good answer to (1), poor answers to (2) and (3).

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Example: Q-learning with a function approximator $\tilde{Q}_w$

$$w \leftarrow w + \alpha \left(y_t - \tilde{Q}_w(s_t, a_t)\right) \nabla_w \tilde{Q}_w(s_t, a_t)$$

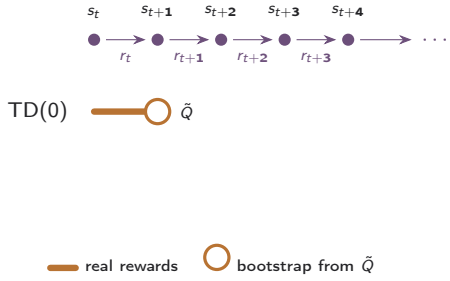
$$y_t = (1-\gamma)\, r_t + \gamma \max_{a'} \tilde{Q}_w(s_{t+1}, a')$$

The max is the greedy step, so evaluate and improve happen together.
The target y_t is one choice of many $\rightarrow$ next slide.

How Is $\tilde{Q}$ Estimated? From TD(0) to Monte Carlo

TD(0): one reward, then bootstrap

$$y_t^{(1)} = (1-\gamma)\, r_t + \gamma\, \tilde{Q}(s_{t+1}, a_{t+1})$$



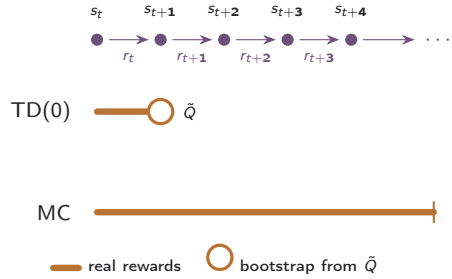
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Monte Carlo: the whole observed return

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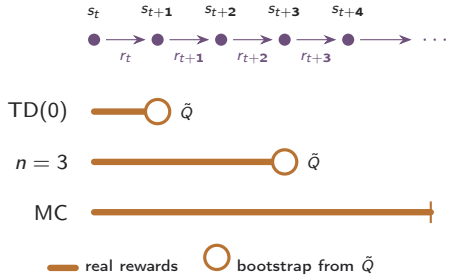
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n-step: in between

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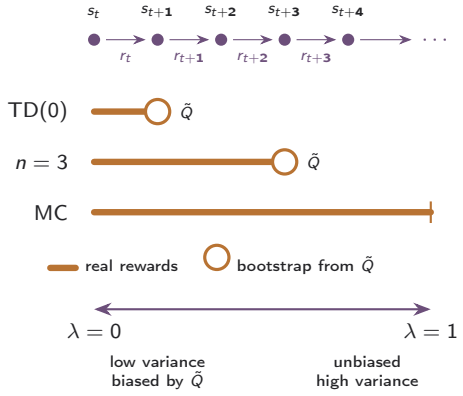
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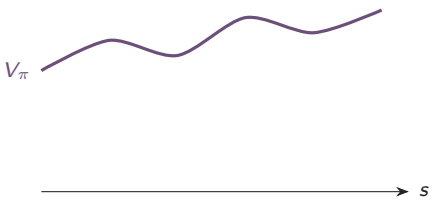
$$y_t^{(n)} = (1-\gamma) \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n\, \tilde{Q}(s_{t+n}, a_{t+n})$$

TD(λ): average over all n

$$y_t^{\lambda} = (1-\lambda) \sum_{n \geq 1} \lambda^{n-1}\, y_t^{(n)}$$



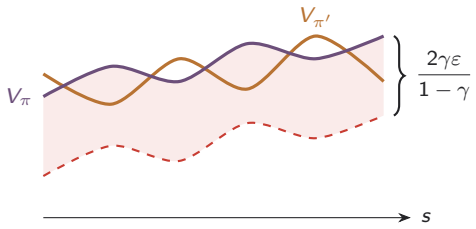
Gap 1: Approximate Value Functions Give No Improvement Guarantee



Approximation error, measured **everywhere** (ℓ_∞):

$$\varepsilon = \max_s \left| \tilde{V}(s) - V_\pi(s) \right|$$

Gap 1: Approximate Value Functions Give No Improvement Guarantee



$V_{\pi'}$ may sit anywhere above the dashed line — **also below** V_π.

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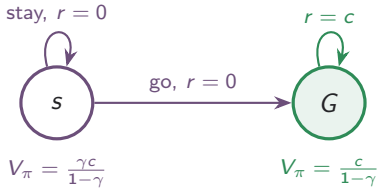
With π' greedy w.r.t. $\tilde{V}$ (Bertsekas & Tsitsiklis):

$$V_{\pi'}(s) \;\geq\; V_\pi(s) - \frac{2\gamma\varepsilon}{1-\gamma} \qquad \forall s$$

Only "won't get **much worse**" — never "gets better".

Gap 1: The Bound Is Tight

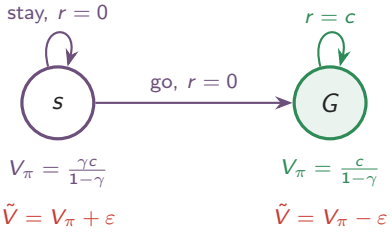
1. π goes to G and collects c forever.



π = go. Set $c = 2\varepsilon$.

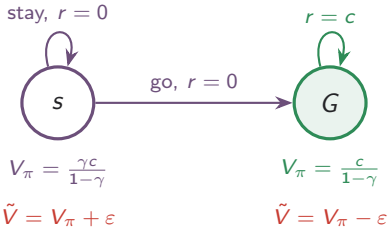
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1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.



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Gap 1: The Bound Is Tight



π = go. Set $c = 2\varepsilon$.

1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.
3. Greedy w.r.t. $\tilde{V}$ at s:

$$\underbrace{\gamma\big(V_\pi(s) + \varepsilon\big)}_{\text{stay}} = \underbrace{\gamma\big(V_\pi(G) - \varepsilon\big)}_{\text{go}}$$

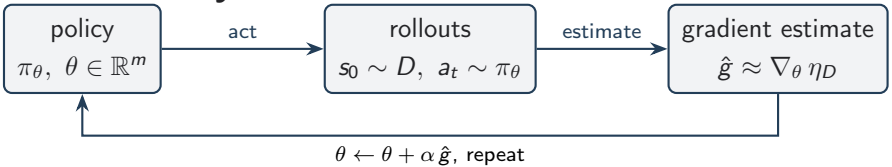
a tie, since $V_\pi(G) - V_\pi(s) = c = 2\varepsilon$, so π' may stay forever: $V_{\pi'}(s) = 0$.

At s: $V_\pi(s) - V_{\pi'}(s) = \dfrac{\gamma c}{1-\gamma} = \dfrac{2\gamma\varepsilon}{1-\gamma}$ — exactly the bound.

At G: one action only, so $\pi' = \pi$ and the loss is 0.

The bound is a max over states; s attains it.

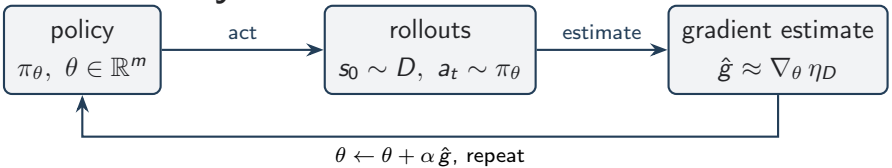
Reminder: Policy Gradient Methods



$$\nabla_\theta \eta_D = \frac{1}{1-\gamma} \sum_{s,a} d_{\pi,D}(s)\, \nabla_\theta \pi(a;s)\, Q_\pi(s,a)$$

Small steps uphill on η_D itself $\Rightarrow$ a good answer to question 1.

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Example: REINFORCE with a Gibbs (softmax) policy

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}, \qquad \hat{g} \ \propto \ \sum_t \nabla_\theta \log \pi_\theta(a_t;s_t)\, \hat{Q}_t$$

$\hat{Q}_t$: return observed after step t. Same policy class as Example 2.

▸ Gibbs policy

Policy Gradient Is Stochastic Gradient Ascent

	Stochastic gradient descent		Policy gradient
objective	$\min_w \; J(w) = \mathbb{E}_X\big[f(w, X)\big]$	$\leftrightarrow$	$\max_\theta \; \eta_D(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\big]$
looking for	$\nabla J(w) = \mathbb{E}\big[\nabla_w f(w, X)\big] = 0$	$\leftrightarrow$	$\nabla_\theta \, \eta_D(\theta) = 0$

Policy Gradient Is Stochastic Gradient Ascent

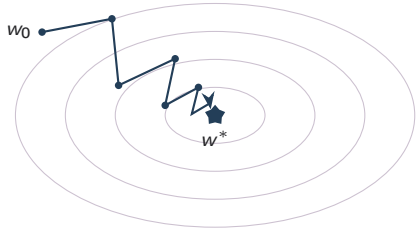
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one sample	$x_k \sim X$	$\leftrightarrow$	rollout τ_k: $s_0 \sim D, \;\; a_t \sim \pi_{\theta_k}$
update	$w_{k+1} = w_k - a_k \, \nabla_w f(w_k, x_k)$	$\leftrightarrow$	$\theta_{k+1} = \theta_k + a_k \, \hat{g}_k$

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update	$w_{k+1} = w_k - a_k \, \nabla_w f(w_k, x_k)$	$\leftrightarrow$	$\theta_{k+1} = \theta_k + a_k \, \hat{g}_k$
noise	$\eta_k = \nabla_w f(w_k, x_k) - \nabla J(w_k)$	$\leftrightarrow$	$\eta_k = \hat{g}_k - \nabla_\theta \, \eta_D(\theta_k)$

What we know about SGD carries over. Twist: samples come from π_θ itself, yet the log-derivative trick (REINFORCE) keeps $\hat{g}_k$ **unbiased**. ▸ Why?

SGD Convergence: the Convex, Smooth Case



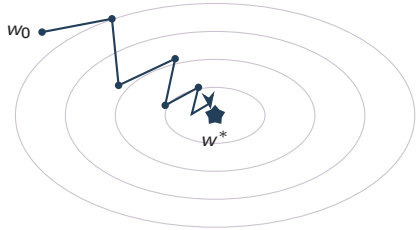
noisy steps, shrinking step size

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\rightarrow$ one minimizer, and bounds on the rate

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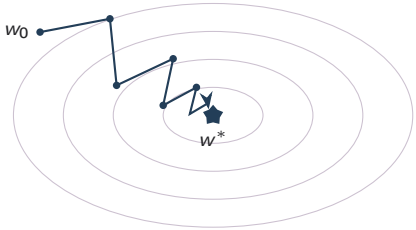
Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\rightarrow$ steps point the right way on average:
the estimate converges to w^*

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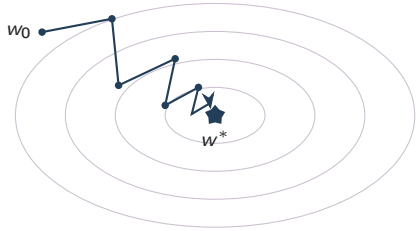
Finite variance and **step sizes**

$$\mathbb{E}[\eta_k^2 \mid \mathcal{H}_k] < \infty$$

$$\sum_k a_k = \infty, \quad \sum_k a_k^2 < \infty$$

$\rightarrow$ steps reach any distance, summed noise stays finite:
noise is filtered out relative to the signal

SGD Convergence: the Convex, Smooth Case



noisy steps, shrinking step size

$w_k \to w^*$ **almost surely**

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\to$ one minimizer, and bounds on the rate

Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\to$ steps point the right way on average:
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Sample Complexity: What Theory Guarantees

$$N(\epsilon, \delta) = \min \left\{ N \in \mathbb{N} \;\middle|\; \mathbb{P}\big(\|w_k - w^*\| \leq \epsilon\big) \geq 1 - \delta \;\text{ for all } k \geq N \right\}$$

samples needed to be ϵ-close with probability $\geq 1 - \delta$ (or in value: $\mathbb{P}(J(w_k) - J(w^*) \leq \epsilon) \geq 1 - \delta$)

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policy gradients with neural nets →

Convex, smooth

global optimum: $N(\epsilon, \delta) = O\big(\sigma^2/(c_1^2 \epsilon^2 \delta)\big)$

Non-convex, smooth

stationary point only: $\mathbb{E}\|\nabla J(w_R)\| \leq \epsilon$ after $K = O(\sigma^2/\epsilon^4)$, w_R: a random iterate, $R \leq K$

Non-convex, non-smooth

weaker still: near-stationary points, special classes

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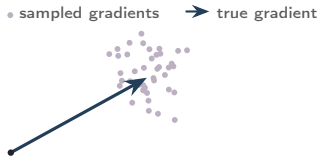
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Non-convex, non-smooth	weaker still: near-stationary points, special classes

Still useful: the convex case shows **what drives** N — the noise σ^2 and the curvature — and that carries over.

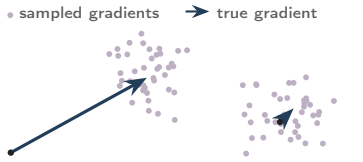
The Signal-to-Noise Reality



large $\|\nabla J\|$:
direction is clear

- Finite variance is enough for convergence *in the limit*.

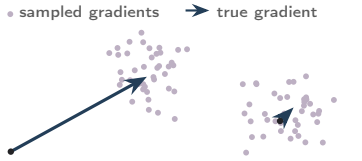
The Signal-to-Noise Reality



- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

$$n \gtrsim \sigma^2/\|\nabla J\|^2$$

The Signal-to-Noise Reality



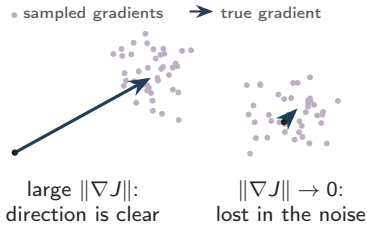
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- **Link to curvature**: strong convexity gives $\|\nabla J(w)\| \geq c_1\|w - w^*\|$, so at distance ϵ:

$$n \gtrsim \sigma^2/(c_1\epsilon)^2 \quad = \text{the convex-row } N \text{ (without } 1/\delta)$$

The Signal-to-Noise Reality



Next: two MDPs with an (almost) zero gradient.

- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

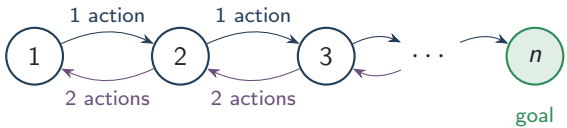
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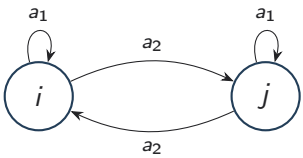
- Flat (c_1 small) or $\|\mathbb{E}[\nabla_w f(w, X)]\| \to 0$: noise dominates, **intractable** n.

Gap 2: Policy Gradients — Example 1 (Exploration)



- Uniform random policy: expected time to goal $= 3(2^n - n - 1) \approx 10^{15}$ for $n = 50$.
- Before reaching the goal, every sensible gradient estimate is 0: right *magnitude*, but **no direction**.
- Importance sampling doesn't help: weights would be exponentially large.

Gap 2: Policy Gradients — Example 2 (Plateaus)

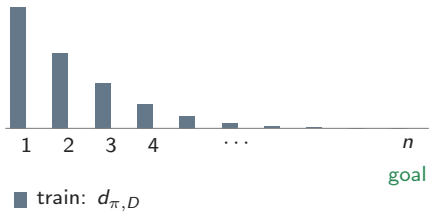


Gibbs policy: $\pi(a; s) \propto \exp(\theta_{sa})$

- Start: $\rho(i) = 0.8,\ \rho(j) = 0.2$.
- Raising the self-loop at i **starves** state j.
- $\rho(j)$ drops to $\approx 10^{-7} \Rightarrow$ nothing is learned at j.
- Result: an extremely **flat plateau** (on a 10^7 time scale).

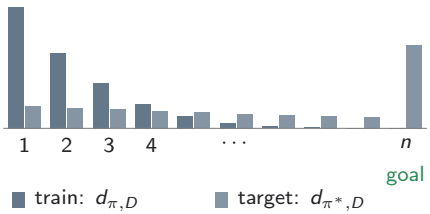
Root cause: η_D weighs improvements by $d_{\pi,D}$ — it is **insensitive to improvement at unlikely states**, even when those states are needed for near-optimal performance.

Gap 2 as Covariate Shift



Train: samples come from where the **current** π goes.

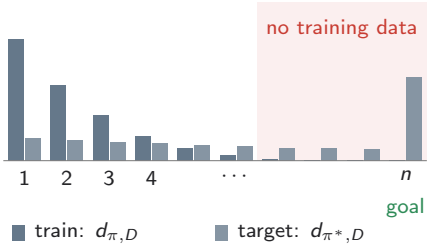
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Gap 2 as Covariate Shift

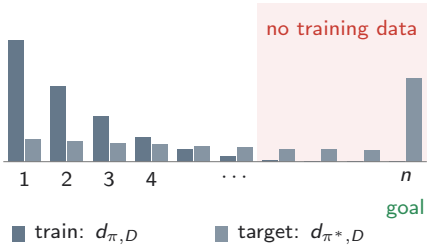


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Gap 2 as Covariate Shift



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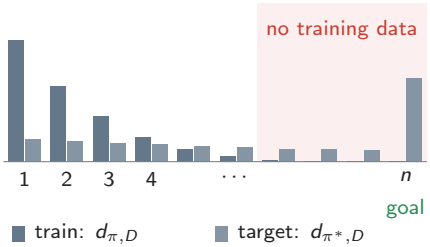
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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Gap 2 as Covariate Shift



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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Instead: change *where the data comes from*: train from μ, still judged by η_D.

$\rightarrow$ restart distribution μ (next slide)

The Two Fixes

1. Conservative update

Don't jump to the greedy policy — mix:

$$\pi_{\text{new}}(a; s) = (1 - \alpha)\,\pi(a; s) + \alpha\,\pi'(a; s)$$

$\alpha \in [0, 1]$ acts as a **step size**. $\alpha = 1$ is greedy policy iteration.

$\rightarrow$ addresses Gap 1 (greedy jumps)

2. Restart distribution μ

Optimize a measure that weighs states more uniformly:

$$\eta_\mu(\pi) = \mathbb{E}_{s\sim\mu}\big[V_\pi(s)\big]$$

μ is "exploratory"; the agent can restart from $s \sim \mu$ (weaker than a generative model*).

D: deployment (given) · μ: training resets (chosen)

$\rightarrow$ addresses Gap 2 (exploration)

- Any optimal policy maximizes both η_μ and η_D — but within a restricted class they can differ.

*Generative model (Kearns et al., 1999): sample $s' \sim P(\cdot \mid s, a)$ at *any* (s, a) you choose. A μ-restart only resets to a *random* $s \sim \mu$. Access, strongest to weakest: full P > generative model > μ-restart > one trajectory, no resets.

Roadmap: From the Two Fixes to the Three Answers

the two fixes

1. Conservative update
mix with a step α

2. Restart distribution
measure η_μ,
restarts $s \sim \mu$

results (next slides)

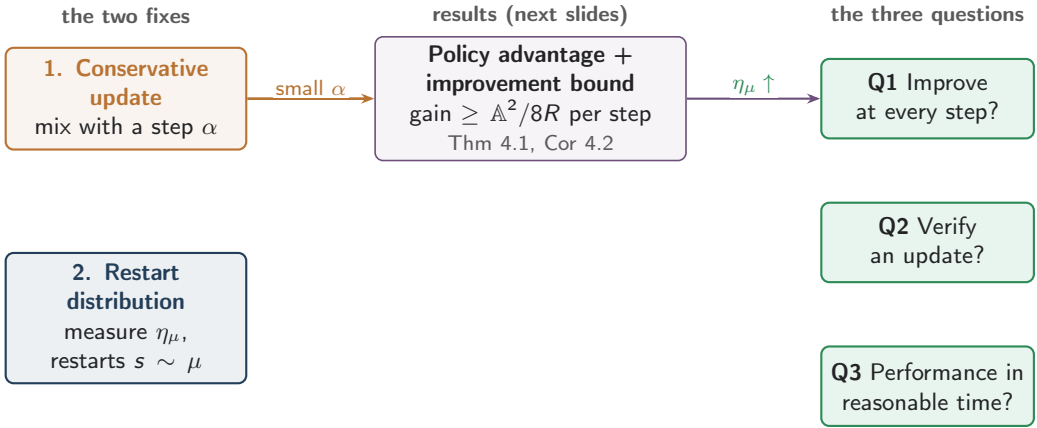
the three questions

Q1 Improve at every step?

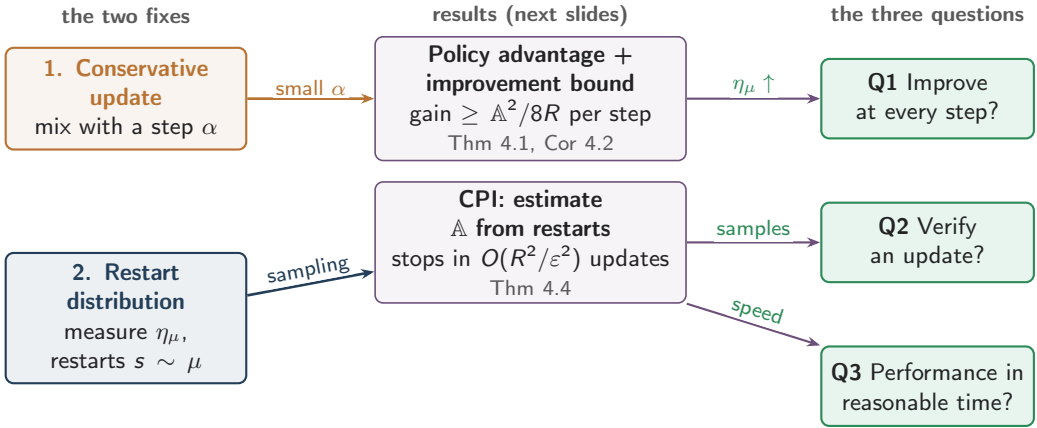
Q2 Verify an update?

Q3 Performance in reasonable time?

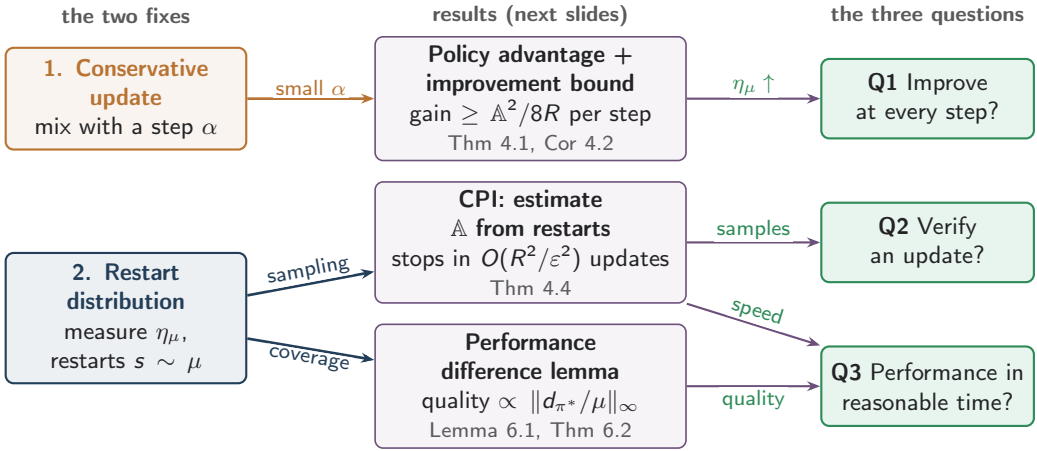
Roadmap: From the Two Fixes to the Three Answers



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Roadmap: From the Two Fixes to the Three Answers



Warm and cool arrows: where each fix enters. μ sets the **quality**, not the **speed**.

The Performance Difference Lemma

Per-state advantage of a policy π' over π — one step of π' at s, then back to π:

$$\bar{A}(s) \;=\; \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Lemma 6.1. For any π, π' and start distribution μ:

▸ Second equality

$$\eta_\mu(\pi') - \eta_\mu(\pi) \;=\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi',\mu}}\big[\bar{A}(s)\big] \;=\; \mathbb{E}\Big[\textstyle\sum_t \gamma^t\,\bar{A}(s_t)\;\Big|\; s_0\sim\mu,\ \pi'\Big]$$

The Performance Difference Lemma

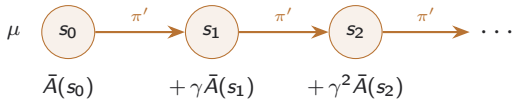
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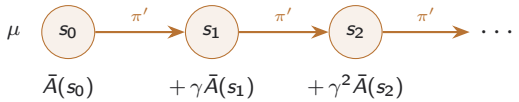
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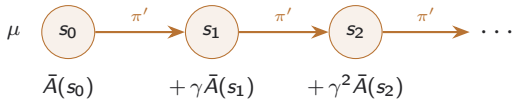
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- **Not measurable:** it needs $d_{\pi',\mu}$, the states of a policy we have not run yet.

A New Quantity: the Policy Advantage

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\bar{A}(s)\big]$$

- Lemma 6.1's average, but **on the states π visits** (from μ) — so it is **measurable**.
- One step of policy improvement maximizes it.
- It is exactly the **first-order** change of η_μ in α:

▸ Proof

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') \qquad \Longrightarrow \qquad \Delta\eta_\mu = \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

Small α + positive policy advantage $\Rightarrow$ improvement. Large $\alpha \Rightarrow$ possible degradation. How large can α be?

The Exact Change vs. the Policy Advantage

Apply Lemma 6.1 to the mixture $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi'$: per state, $\mathbb{E}_{a\sim\pi_{\text{new}}} A_\pi(s,a) = \alpha\,\bar{A}(s)$.

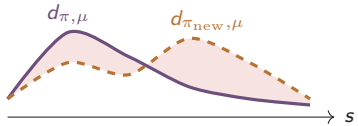
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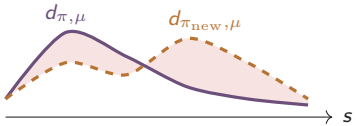
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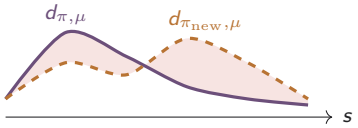
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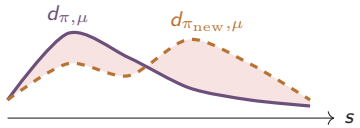
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- $\alpha\mathbb{A}$: the first-order term from the policy-advantage slide.
- Gap: two factors of α — smaller $\bar{A}$ **and** less shift $\Rightarrow O(\alpha^2)$.
- Thm 4.1 = a **lower bound on the gap**.

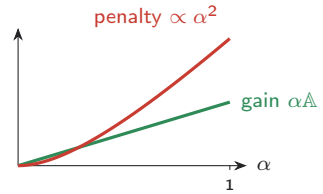
Exact but unmeasurable (Lemma 6.1) = measurable ($\alpha\mathbb{A}$) + gap. We only need to bound the gap.

How Big a Step? The Improvement Bound

Theorem 4.1. With $\varepsilon = \max_s \left|\mathbb{E}_{a\sim\pi'}[A_\pi(s,a)]\right|$, for all $\alpha \in [0,1]$:

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\geq\; \frac{\alpha}{1-\gamma}\left(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\right)$$

- Gain $\alpha\mathbb{A}$: **linear**, measured on old states.
- Penalty: worst case of the state shift, $\propto \alpha^2$.
- Small α: the gain wins $\Rightarrow$ **guaranteed progress**.
- $\alpha = 1$: B&T penalty **plus a gain**. ▸ Compare



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$. Green: improvement guaranteed.

Some small step always helps when $\mathbb{A} > 0$. **Which** α? $\rightarrow$ next slide.

Choosing the Step: Corollary 4.2

1. **Simplify:** lower the bound further, using $|A_\pi| \le R$, so that only the measurable $\mathbb{A}$ is left.

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\Big) \qquad \text{Thm 4.1}$$

$$\;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha R}{1-\gamma}\Big) \;=:\; f(\alpha) \qquad \varepsilon \le R,\;\; 1-\gamma(1-\alpha) \ge 1-\gamma,\;\; \gamma \le 1$$

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2. Maximize: $f(\alpha) = \frac{\mathbb{A}}{1-\gamma}\,\alpha - \frac{2R}{(1-\gamma)^2}\,\alpha^2$ is a downward parabola in α.

$$f'(\alpha) \;=\; \frac{\mathbb{A}}{1-\gamma} - \frac{4R\,\alpha}{(1-\gamma)^2} \;=\; 0 \quad\Longrightarrow\quad \alpha^* \;=\; \frac{(1-\gamma)\mathbb{A}}{4R}$$

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3. Plug in:

$$f(\alpha^*) \;=\; \frac{\mathbb{A}}{1-\gamma}\cdot\frac{(1-\gamma)\mathbb{A}}{4R} \;-\; \frac{2R}{(1-\gamma)^2}\cdot\frac{(1-\gamma)^2\mathbb{A}^2}{16R^2} \;=\; \frac{\mathbb{A}^2}{8R}$$

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

Choosing the Step: Corollary 4.2

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

Computable: $\mathbb{A}$ averages over the *old* policy's states (μ-restarts); R is known. Price: $\mathbb{A} \le R$ gives $\alpha \le (1-\gamma)/4$ (≤ 0.0025 at $\gamma = 0.99$), and $\alpha \to 0$ as $\mathbb{A} \to 0$: **tiny steps**.

Conservative Policy Iteration

1. $\pi' \leftarrow G_\varepsilon(\pi, \mu)$ (greedy policy chooser)
2. Estimate $\hat{\mathbb{A}} \approx \mathbb{A}_{\pi,\mu}(\pi')$ to $\pm\frac{\varepsilon}{3}$ with $O\big(\frac{R^2}{\varepsilon^2} \log \frac{R^2}{\delta\varepsilon^2}\big)$ μ-restarts
3. If $\hat{\mathbb{A}} < \frac{2\varepsilon}{3}$: **stop**, return π
4. Else: mix with $\alpha = \frac{(1-\gamma)(\hat{\mathbb{A}} - \varepsilon/3)}{4R}$, go to 1

G_ε: returns π' with $\mathbb{A}_{\pi,\mu}(\pi') \geq \max_{\pi''} \mathbb{A}_{\pi,\mu}(\pi'') - \varepsilon$ — e.g. a regression on advantages with *average* error $\leq \varepsilon/2$.

▸ How?

Theorem 4.4. With prob. $\geq 1 - \delta$, CPI:

i) improves η_μ at **every** update;

ii) stops within $72\, R^2/\varepsilon^2$ calls to G_ε;

iii) returns π with $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$.[1]

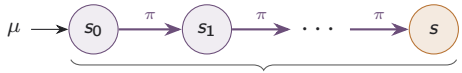
Number of updates does **not** depend on μ or on the state-space size.

[1] $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) \equiv \max_{\pi'} \mathbb{A}_{\pi,\mu}(\pi') = \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\max_a A_\pi(s,a)\big]$: the largest policy advantage any π' can achieve (Def. 4.3). Small $\Rightarrow$ no policy improves much on π under $d_{\pi,\mu}$.

Verifying an Update: Estimating $\mathbb{A}$ From μ-Restarts

Since $V_\pi(s) = \sum_a \pi(a;s)\, Q_\pi(s,a)$, the policy advantage needs only Q_π:

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

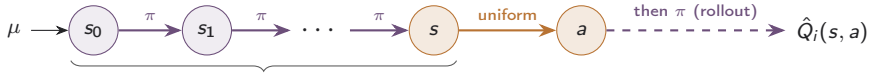


stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

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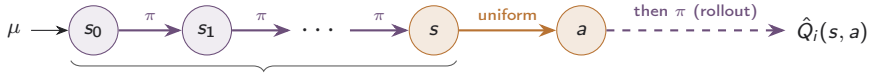


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stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

$$x_i = n_a\, \hat{Q}_i(s,a)\big(\pi'(a;s) - \pi(a;s)\big) \in [-n_a R,\, n_a R], \qquad \hat{\mathbb{A}} = \tfrac{1}{k}\sum_{i=1}^{k} x_i$$

$$\text{Hoeffding:} \quad \Pr\big(|\hat{\mathbb{A}} - \mathbb{A}| > \Delta\big) \;\le\; 2\, e^{-k\Delta^2/(2n_a^2R^2)}$$

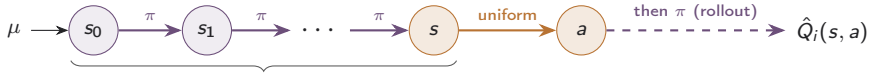
- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, [2]) ▸ IS ▸ Hoeffding

[2] n_a is the number of actions.

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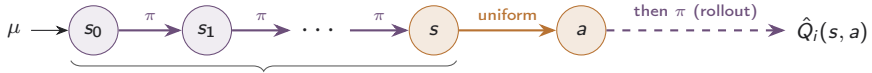
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- Only rollouts of the **current** π — no need to run π_{new}. This is **Q2**.

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stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

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- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, ▸ IS)[2] ▸ Hoeffding
- Only rollouts of the **current** π — no need to run π_{new}. This is **Q2**.
- CPI: $\Delta = \varepsilon/3$, so $\hat{\mathbb{A}} - \frac{\varepsilon}{3} \le \mathbb{A}$ is safe for the stop test and for α.

[2] n_a is the number of actions.

How Good Is the Policy Found?

Theorem 6.2. If $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < \varepsilon$, then for any $\tilde{\mu}$:

$$\eta_{\tilde{\mu}}(\pi^*) - \eta_{\tilde{\mu}}(\pi) \;\le\; \frac{\varepsilon}{1-\gamma} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{d_{\pi,\mu}} \right\|_\infty \;\le\; \frac{\varepsilon}{(1-\gamma)^2} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{\mu} \right\|_\infty$$

- **Covariate-shift bound:** test risk $\le$ density ratio $\times$ train risk.
  - train risk: $\mathrm{OPT}(\mathbb{A}_{\pi,\mu})$, on CPI's data $d_{\pi,\mu}$
  - test: the states π^* visits, $d_{\pi^*,\tilde{\mu}}$
- **Uniform μ caps the ratio**: $d_{\pi,\mu} \ge (1-\gamma)\mu \Rightarrow$ ratio $\le |S|/(1-\gamma)$, for any π^*.

CPI: Answers to the Three Questions

1. **Improving measure:** η_μ improves at every update (Thm. 4.4).
2. **Verification:** estimate the policy advantage $\mathbb{A}$ from samples (Hoeffding).
3. **Performance:** within $O(R^2/\varepsilon^2)$ updates, close to optimal up to the mismatch factor (Thm. 6.2).

Kept going forward

- Surrogate on the old state distribution
- Improvement lower bound = gain − penalty
- Small, controlled steps

Limitations

- Only **mixture** policies
- Needs a restart distribution
- Theoretical step sizes are tiny

Like non-convex SGD, but Thm. 6.2 turns "stationary" into "near-optimal".

▸ CPI vs. SGD

Trust Region Policy Optimization

Schulman, Levine, Moritz, Jordan & Abbeel
ICML 2015

Introduces **TRPO**: CPI's guarantee for **any** stochastic policy π_θ

Setting & Assumptions

An MDP $(\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$:

- **Finite** $\mathcal{S}, \mathcal{A}$ in the proofs — also holds for **continuous** (conditions later).
- **Discounted**: $\gamma \in (0, 1)$, reward $r(s)$.
- **Unknown dynamics**: sample trajectories only.
- **Start distribution** ρ_0 (CPI's D) — **no restart** μ.
- **Unnormalized**: $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$, $s_0 \sim \rho_0$.
- **Any** differentiable stochastic policy π_θ (e.g. a neural net).

TRPO: Starting From the Same Identity

Unnormalized return $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$ and visitation $\rho_\pi(s) = \sum_t \gamma^t P(s_t = s)$.

$$\eta(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a) \qquad \text{(performance difference)}$$

$\rho_{\tilde{\pi}}$ depends on $\tilde{\pi}$ in a complicated way $\Rightarrow$ replace it by the **old** visitation:

$$L_\pi(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_\pi(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a)$$

- Matches η to first order: $L_{\pi_{\theta_0}}(\pi_{\theta_0}) = \eta(\pi_{\theta_0})$ and equal gradients at θ_0.
- Same idea as CPI's policy advantage — but says nothing about **how big** a step is safe.

From Mixtures to Any Stochastic Policy

CPI's bound only holds for mixtures. TRPO replaces the mixing weight α by a **distance** between policies:

$$\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}}) = \max_s D_{\mathrm{TV}}\big(\pi_{\mathrm{old}}(\cdot \mid s) \,\|\, \pi_{\mathrm{new}}(\cdot \mid s)\big)$$

Theorem 1.

$$\eta(\pi_{\mathrm{new}}) \;\geq\; L_{\pi_{\mathrm{old}}}(\pi_{\mathrm{new}}) - \frac{4\epsilon\gamma}{(1-\gamma)^2}\,\alpha^2, \qquad \epsilon = \max_{s,a} |A_\pi(s,a)|$$

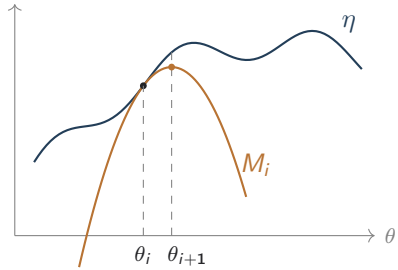
- Proof idea: two distributions with $D_{\mathrm{TV}} \leq \alpha$ can be **coupled** to agree w.p. $1-\alpha$ — so the new policy "acts like a mixture".
- Note the different ϵ: max over (s,a) instead of CPI's max over s of an expectation.

From TV to KL: a Minorize–Maximize Algorithm

Use $D_{\mathrm{TV}}(p\|q)^2 \le D_{\mathrm{KL}}(p\|q)$:

$$\eta(\tilde{\pi}) \ge \underbrace{L_\pi(\tilde{\pi}) - C\, D_{\mathrm{KL}}^{\max}(\pi, \tilde{\pi})}_{M_i(\tilde{\pi})}, \quad C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$$

- $M_i \le \eta$ everywhere, with equality at π_i.
- Maximize $M_i \Rightarrow$ $\eta(\pi_{i+1}) - \eta(\pi_i) \ge M_i(\pi_{i+1}) - M_i(\pi_i) \ge 0$.
- **Monotone improvement** (with exact advantages).



Continuous States and Actions?

Back to footnote 1: "replace the sums with integrals". **True** — no proof uses finiteness.

Why it carries over

- **Perf. difference**: telescoping, any space.
- **Coupling**: mass $1 - D_{\mathrm{TV}}$ on $\{a = \tilde{a}\}$ — also for densities.
- **Perturbation** (App. B): matrices $\to$ Markov operators, same norms.

What "integrals" hides

1. **Bounded** $\epsilon = \sup_{s,a} |A_\pi(s,a)| < \infty$.*
2. **max** $\to$ **sup** over **all** states, not just visited.
3. **Stochastic** policies: deterministic $\Rightarrow D_{\mathrm{KL}} = \infty$.
4. **Measurable** kernels; ρ_π is a measure.

* Often **vacuous** in continuous control: Gaussian actions with reward $-\|u\|^2$ (their Swimmer) give $\epsilon = \infty$. The practical algorithm drops the bound anyway (mean KL, fixed δ).

From Penalty to Constraint: the Trust Region

Theory (penalty)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) - C\, D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta)$$

With the theoretical C, steps are **tiny**; and a good C is hard to choose robustly.

Practice (constraint)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta)$$

$$\text{s.t.} \;\; D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \leq \delta$$

δ is in policy space, so it transfers across problems ($\delta = 0.01$ in all experiments).

The guarantee is now heuristic: we optimize a surrogate inside a region where we *trust* it.

From Max KL to Mean KL

- $D_{\mathrm{KL}}^{\max}$ is one constraint **per state** — impractical to enforce or estimate.
- Replace it with the average under the old visitation distribution:

$$\bar{D}_{\mathrm{KL}}^{\rho}(\theta_1, \theta_2) = \mathbb{E}_{s\sim\rho}\Big[D_{\mathrm{KL}}\big(\pi_{\theta_1}(\cdot \mid s) \,\|\, \pi_{\theta_2}(\cdot \mid s)\big)\Big]$$

$$\max_{\theta} \; L_{\theta_{\mathrm{old}}}(\theta) \quad \text{s.t.} \quad \bar{D}_{\mathrm{KL}}^{\rho_{\theta_{\mathrm{old}}}}(\theta_{\mathrm{old}}, \theta) \le \delta$$

- An **expectation** $\Rightarrow$ can be estimated from samples.
- Empirically performs like the max-KL version.

Turning Sums Into Samples

Expand $L_{\theta_{\text{old}}}$ and apply three substitutions:

$$\max_{\theta} \sum_{s} \rho_{\theta_{\text{old}}}(s) \sum_{a} \pi_{\theta}(a \mid s)\, A_{\theta_{\text{old}}}(s, a)$$

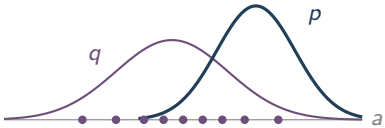
1. $\sum_s \rho_{\theta_{\text{old}}}(s)[\cdot] \;\rightarrow\; \frac{1}{1-\gamma}\mathbb{E}_{s\sim\rho_{\theta_{\text{old}}}}[\cdot]$
2. $A_{\theta_{\text{old}}} \;\rightarrow\; Q_{\theta_{\text{old}}}$ (changes the objective by a constant)
3. $\sum_a \;\rightarrow$ **importance sampling** with a sampling distribution q

$$\max_{\theta} \; \mathbb{E}_{s\sim\rho_{\theta_{\text{old}}},\, a\sim q}\left[\frac{\pi_{\theta}(a \mid s)}{q(a \mid s)}\, Q_{\theta_{\text{old}}}(s, a)\right]$$
$$\text{s.t.} \;\; \mathbb{E}_{s\sim\rho_{\theta_{\text{old}}}}\left[D_{\text{KL}}(\pi_{\theta_{\text{old}}}(\cdot \mid s) \,\|\, \pi_{\theta}(\cdot \mid s))\right] \leq \delta$$

Importance Sampling: Reweighting Samples

Weight $w = p/q$ (needs $q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$



Samples come from q, but we want an average under p.

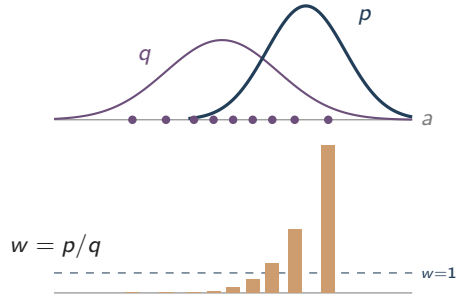
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$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$



p far from q: one rare sample carries a huge weight $\Rightarrow$ high variance.

Importance Sampling: Reweighting Samples

Weight $w = p/q$ (needs $q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

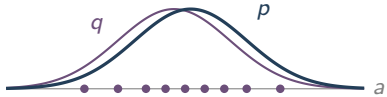
Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$

In TRPO (single path), at each state s

$$p = \pi_\theta(\cdot \mid s), \quad q = \pi_{\theta_{\text{old}}}(\cdot \mid s)$$

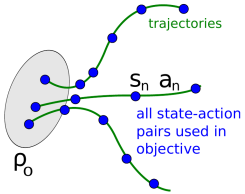
$$w = \frac{\pi_\theta(a \mid s)}{\pi_{\theta_{\text{old}}}(a \mid s)}$$

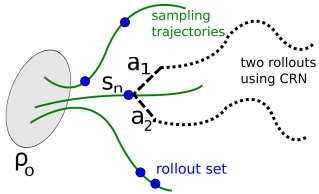




Trust region keeps $\pi_\theta \approx \pi_{\theta_{\text{old}}} \Rightarrow$ all $w \approx 1$.
Only actions are reweighted; states stay $s \sim \rho_{\theta_{\text{old}}}$.

Two Sampling Schemes





Single path

- Roll out $\pi_{\theta_{\text{old}}}$; use **every** (s_n, a_n).
- $q = \pi_{\theta_{\text{old}}}$; $\hat{Q}$ = discounted return-to-go.
- **No state resets** — works on real systems.

Vine

- From each s_n in the **rollout set**: try K actions $a_k \sim q$, short rollout each (CRN).
- **Lower variance** $\hat{Q}$.
- Needs **resets** to arbitrary states — like CPI's μ-restarts.

Solving the Trust-Region Problem (1): Search Direction

Approximate around θ_{old} — **linear** objective, **quadratic** constraint:

$$L(\theta) \approx L(\theta_{\text{old}}) + g^\top(\theta - \theta_{\text{old}}), \qquad \bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \approx \tfrac{1}{2}(\theta - \theta_{\text{old}})^\top A\,(\theta - \theta_{\text{old}})$$

- $A_{ij} = \frac{\partial^2}{\partial\theta_i \partial\theta_j}\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta)$ is the **Fisher information matrix**.
- Solution direction: $s \propto A^{-1}g$ — the **natural gradient**.

A is computed *analytically* as the Hessian of the KL (integrating over actions), not as the covariance of sampled score functions.

Solving the Trust-Region Problem (2): Conjugate Gradient

- Forming or inverting A (size $|\theta| \times |\theta|$) is impossible for neural networks.
- **Conjugate gradient** solves $Ax = g$ using only **Fisher-vector products** $y \mapsto Ay$.

Fisher-vector product. If the network outputs distribution parameters $\mu_\theta(x)$:

$$A = J^\top M J, \qquad J = \frac{\partial \mu}{\partial \theta}, \quad M = \text{Fisher of the distribution w.r.t. } \mu$$

Jy and $J^\top v$ are standard autodiff operations (forward / backprop).

- $k = 10$ CG iterations is enough.
- Compute Fisher-vector products on $\sim$10% of the data $\Rightarrow$ total cost $\approx$ one gradient.

Solving the Trust-Region Problem (3): Step Size & Line Search

- Given direction $s \approx A^{-1}g$, take the largest step that hits the KL boundary:

$$\delta = \tfrac{1}{2}(\beta s)^\top A(\beta s) \quad \Longrightarrow \quad \beta = \sqrt{\frac{2\delta}{s^\top A s}}$$

- The approximations are only local, so do a **backtracking line search**: shrink β exponentially until
  - the true surrogate $L_{\theta_{\text{old}}}(\theta)$ improves, **and**
  - the true constraint $\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \leq \delta$ holds.

Without the line search, TRPO occasionally takes large steps that cause **catastrophic** performance drops.

The TRPO Algorithm

Repeat:

1. Collect (s, a) pairs with Monte Carlo $\hat{Q}$ estimates (single path or vine).
2. Build the sample estimates of the surrogate L and the mean-KL constraint.
3. Compute $g = \nabla_\theta L$; solve $As = g$ with conjugate gradient (Fisher-vector products).
4. Step size $\beta = \sqrt{2\delta / s^\top As}$; backtracking line search on L and the KL.
5. $\theta_{\text{old}} \leftarrow \theta_{\text{old}} + \beta s$.

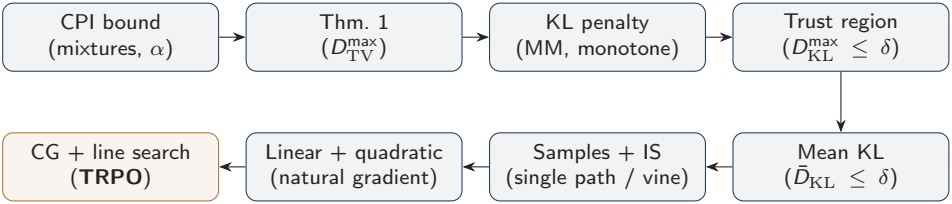
Cost per update: only slightly more than computing the gradient itself.

One Problem, Several Known Algorithms

Objective	Constraint / penalty	Algorithm
Linear in θ	$\frac{1}{2}\|\theta - \theta_{\text{old}}\|^2 \leq \delta$	Vanilla policy gradient
Linear in θ	Quadratic KL, fixed step $\frac{1}{\lambda}$	Natural policy gradient
Full $L_{\pi_{\text{old}}}(\pi)$	none	Policy iteration
$L_{\theta_{\text{old}}}(\theta)$	$\bar{D}_{\text{KL}} \leq \delta$, enforced	**TRPO**

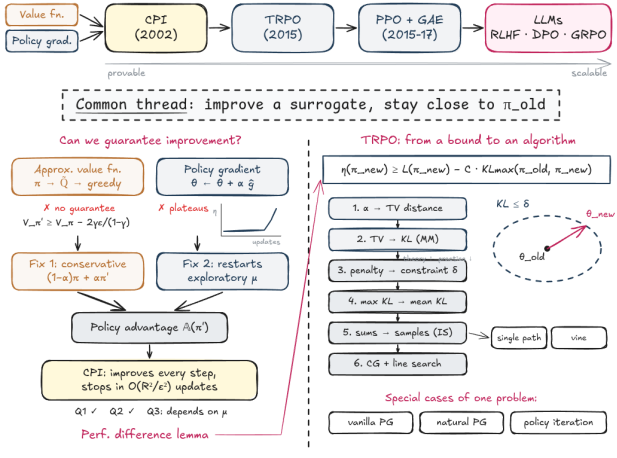
- NPG vs. TRPO: same direction $A^{-1}g$; TRPO **enforces** the KL constraint each step — matters on larger problems.
- Related: REPS constrains the joint $p(s, a)$; TRPO constrains only $\pi(a \mid s)$.

Recap: From Theory to Practice



- Each arrow trades a guarantee for practicality.
- What's left: a heavy second-order solver $\Rightarrow$ motivation for simpler, first-order methods.

Policy Optimization: From Guarantees to LLMs



Part II — Main Sources



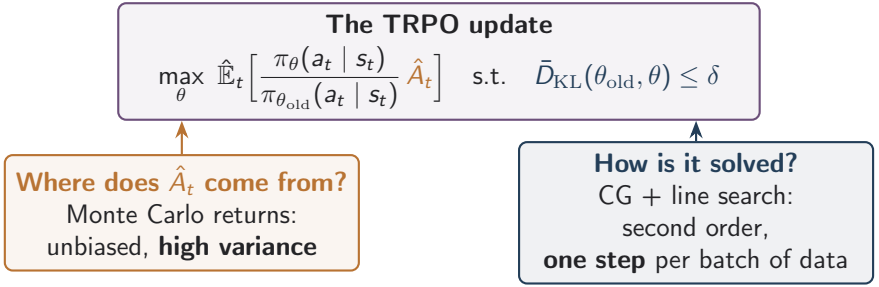
Published parallel to TRPO, this introduced the exponentially-weighted advantage estimator used inside almost all modern PPO and TRPO implementations to balance variance and bias.

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l (r_{t+l} + \gamma V(s_{t+l+1}) - V(s_{t+l}))$$

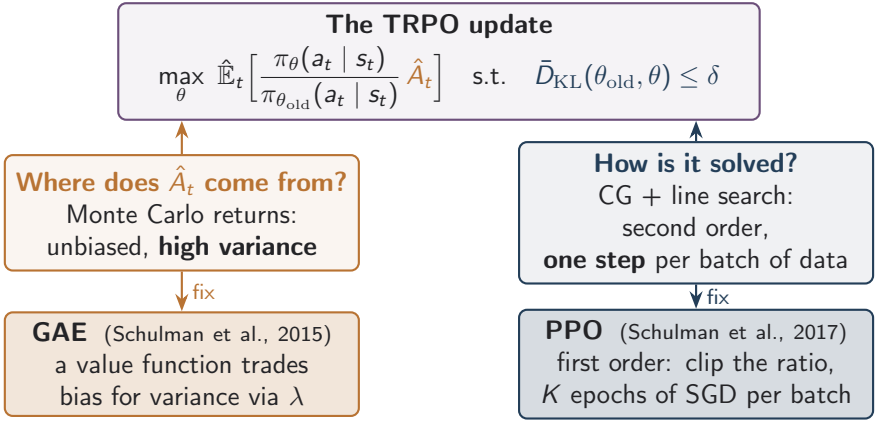


Simplified TRPO by replacing the expensive KL penalty with a clipped surrogate objective function. This became the industry standard for Reinforcement Learning from Human Feedback (RLHF).

Where TRPO Leaves Us



Where TRPO Leaves Us



High-Dimensional Continuous Control Using Generalized Advantage Estimation

Schulman, Moritz, Levine, Jordan & Abbeel
ICLR 2016 (arXiv 2015)

Introduces **GAE**(γ, λ): TD(λ) for advantages

Setting & Assumptions

- Same MDP as TRPO: start distribution ρ_0, unknown dynamics, reward $r_t = r(s_t, a_t, s_{t+1})$.
- **Undiscounted** goal: maximize $\mathbb{E}\big[\sum_t r_t\big]$ (assumed finite).
- γ is **not** part of the problem: an algorithm knob that trades bias for variance.
- Any policy-gradient method — here TRPO for the policy, a trust region for V_ϕ too.
- 3D robots, neural nets with $> 10^4$ parameters, torques from raw kinematics.

Which Weight Multiplies the Score Function?

$$g = \mathbb{E}\Big[\sum_{t\geq 0} \Psi_t \, \nabla_\theta \log \pi_\theta(a_t \mid s_t)\Big]$$

Same expectation for each choice of Ψ_t — **different variance**:

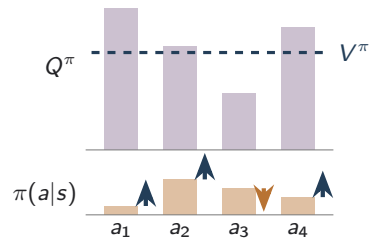
1	$\sum_{t'\geq 0} r_{t'}$	total return
2	$\sum_{t'\geq t} r_{t'}$	reward after a_t
3	$\sum_{t'\geq t} r_{t'} - b(s_t)$	with a baseline
4	$Q^\pi(s_t, a_t)$	action value
5	$A^\pi(s_t, a_t)$	advantage
6	$r_t + V^\pi(s_{t+1}) - V^\pi(s_t)$	TD residual

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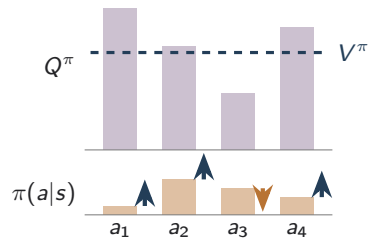
Step along $A \nabla \log \pi$: raise $\pi(a|s)$ **iff** a beats the average V^π.

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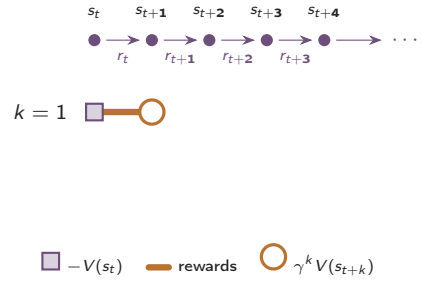
Step along $A \nabla \log \pi$: raise $\pi(a|s)$ **iff** a beats the average V^π.

Discount as a knob: $A^{\pi,\gamma}$ ignores effects $\gg \frac{1}{1-\gamma}$ steps later $\Rightarrow$ less variance, **some bias**.

GAE: TD(λ) for Advantages

TD residual: one reward, then V

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$



GAE: TD(λ) for Advantages

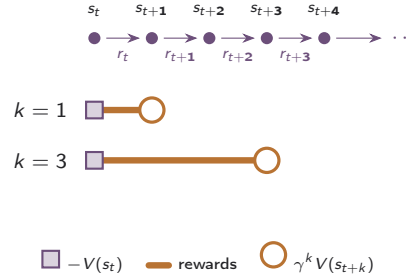
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k residuals: $\hat{A}_t^{(k)} = \sum_{l<k} \gamma^l \delta_{t+l}^V$ telescopes to

$$\hat{A}_t^{(k)} = -V(s_t) + \sum_{l=0}^{k-1} \gamma^l r_{t+l} + \gamma^k V(s_{t+k})$$



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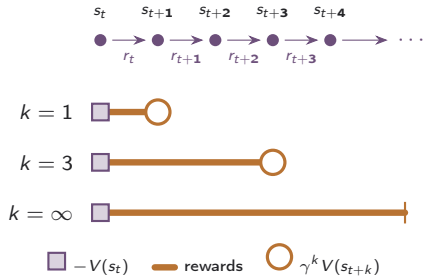
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all of them: return minus a baseline

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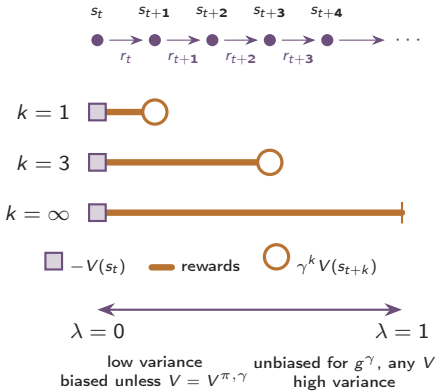
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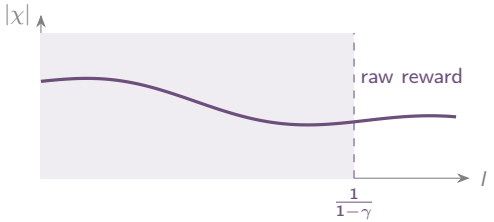
$$\hat{A}_t^{(\infty)} = \sum_{l \geq 0} \gamma^l r_{t+l} - V(s_t)$$

GAE: average all k with weights $(1-\lambda)\lambda^{k-1}$

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l \geq 0} (\gamma\lambda)^l \delta_{t+l}^V$$



Two Knobs, Two Kinds of Bias



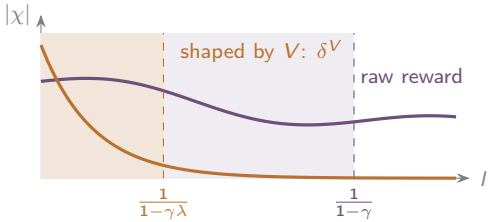
$\chi(l; s, a) = \mathbb{E}[r_{t+l} \mid s, a] - \mathbb{E}[r_{t+l} \mid s]$: how long a_t keeps mattering

γ: cut the horizon

Drops effects after $\approx \frac{1}{1-\gamma}$ steps.

Biased **even with the exact** V.

Two Knobs, Two Kinds of Bias

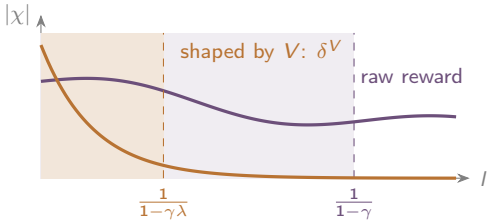


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δ^V is the reward shaped by $\Phi = V$: a good V pulls the effect of a_t to $l \approx 0$.
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3D biped: best $\gamma \in [0.99, 0.995]$, $\lambda \in [0.96, 0.99]$.
$\lambda = 0$ (plain TD residual): too biased.

GAE + TRPO: Policy and Value Function

Repeat:

1. Run π_{θ_i} for N timesteps.
2. δ_t^V with the **current** $V = V_{\phi_i}$.
3. $\hat{A}_t = \sum_l (\gamma\lambda)^l \delta_{t+l}^V$.
4. Policy: TRPO step with $\hat{A}_t \to \theta_{i+1}$.
5. Value: trust-region fit $\to \phi_{i+1}$.

Order matters: an overfit V makes every $\delta_t^V = 0$, so the gradient would vanish.

Value fit with a trust region

$$\min_{\phi} \sum_n \left\| V_\phi(s_n) - \hat{V}_n \right\|^2$$

$$\text{s.t.} \quad \frac{1}{N} \sum_n \frac{\| V_\phi(s_n) - V_{\phi_{\text{old}}}(s_n) \|^2}{2\sigma^2} \leq \epsilon$$

$=$ a KL trust region for a Gaussian with mean $V_\phi(s)$; solved with CG, like TRPO.

3D biped learns to run from raw kinematics: 1000 iterations $\approx$ 5.8 days of simulated time.

Proximal Policy Optimization Algorithms

Schulman, Wolski, Dhariwal, Radford & Klimov
arXiv 2017 (OpenAI)

Introduces **PPO**: TRPO's reliability with only **first-order** SGD

Setting & Motivation

- Same setting as TRPO; advantages $\hat{A}_t$ from (truncated) **GAE**.
- **TRPO**: complicated; no dropout, no parameter sharing between policy and value.
- **Vanilla PG**: several steps on the same batch $\Rightarrow$ destructively large updates.
- **Goal**: reuse each batch for many SGD steps, without leaving the trust region.
- Tested on MuJoCo and 49 Atari games.

Back to the CPI Surrogate

Ratio = TRPO's importance weight: $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \quad r_t(\theta_{\text{old}}) = 1$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t\left[r_t(\theta)\,\hat{A}_t\right] \qquad \text{("CPI": Kakade \& Langford)}$$

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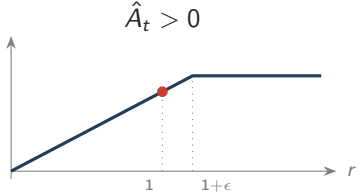
Constraint (TRPO)
s.t. $\hat{\mathbb{E}}_t[D_{\text{KL}}] \le \delta$
second-order solver

Penalty
$-\beta\,\hat{\mathbb{E}}_t[D_{\text{KL}}]$
no single β works $\rightarrow$ adapt it

Clip (PPO)
no gain once $|r_t - 1| > \epsilon$
first order, next slide

The Clipped Surrogate Objective

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min\left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\right)\Big], \qquad \epsilon = 0.2$$

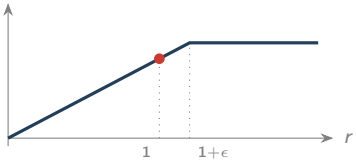


Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

The Clipped Surrogate Objective

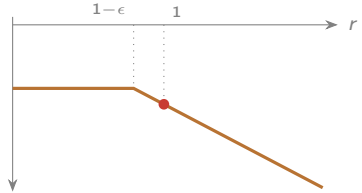
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$\hat{A}_t > 0$



Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

$\hat{A}_t < 0$

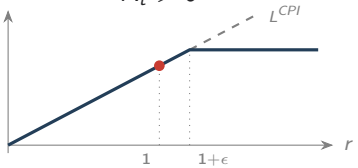


Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.

The Clipped Surrogate Objective

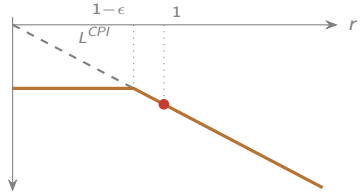
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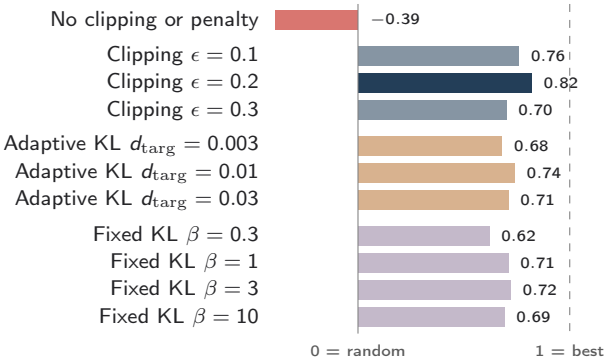
$\hat{A}_t < 0$



Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.

min $\Rightarrow$ $L^{CLIP} \leq L^{CPI}$: clipping removes **gains**, never hides **losses** — a pessimistic bound.

Clipping vs. KL Penalty: the Ablation



Adaptive KL penalty

$$L^{KLPEN} = \hat{\mathbb{E}}_t\left[r_t \hat{A}_t - \beta\, D_{\text{KL}}[\pi_{\theta_{\text{old}}}, \pi_\theta]\right]$$

After each update, with $d = \hat{\mathbb{E}}_t[D_{\text{KL}}]$:

- $d < d_{\text{targ}}/1.5$: $\beta \leftarrow \beta/2$
- $d > 1.5\, d_{\text{targ}}$: $\beta \leftarrow 2\beta$

Clipping with $\epsilon = 0.2$ wins; the KL penalty stays a useful baseline.

PPO in Practice

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t\Big[\underbrace{L_t^{CLIP}(\theta)}_{\text{policy}} - c_1\underbrace{\big(V_\theta(s_t) - V_t^{\text{targ}}\big)^2}_{\text{value}} + c_2\underbrace{S[\pi_\theta](s_t)}_{\text{entropy}}\Big]$$

Each iteration:

1. N actors run $\pi_{\theta_{\text{old}}}$ for T steps each.
2. Truncated GAE: $\hat{A}_t = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l\delta_{t+l}$.
3. K epochs of minibatch Adam on L.
4. $\theta_{\text{old}} \leftarrow \theta$.

MuJoCo settings

horizon T	2048
epochs K	10
minibatch	64
$\gamma,\ \lambda$	0.99, 0.95
clip ϵ	0.2

K epochs on the same batch is the point: TRPO took **one** step per batch.

TRPO vs. PPO

	TRPO	PPO
Stay close by	KL constraint $\bar{D}_{\mathrm{KL}} \leq \delta$	clipped ratio $[1-\epsilon, 1+\epsilon]$
Solver	CG + line search (2nd order)	minibatch Adam (1st order)
Steps per batch	one	K epochs
Policy + value	separate networks	may share (one loss)
Advantages	Monte Carlo $\hat{Q}$	truncated GAE

Kept from CPI and TRPO

- Surrogate on the **old** samples
- The ratio $\pi_\theta / \pi_{\theta_{\mathrm{old}}}$
- Small, controlled steps

Given up

- Any improvement **bound**
- An exact limit on the KL
- (Monotone improvement: never had it in practice)

Part III — Main Sources

Deep RL from Human Preferences

Christiano et al. (2017)

The foundational paper for modern RLHF. It established how to train a separate neural network reward model from human pairwise comparisons of trajectory segments (σ^1, σ^2), which is then used to optimize the policy. The reward model optimizes the Bradley-Terry preference probability:

$$\hat{P}[\sigma^1 \succ \sigma^2] = \frac{\exp \sum_t \hat{r}(o_t^1, a_t^1)}{\exp \sum_t \hat{r}(o_t^1, a_t^1) + \exp \sum_t \hat{r}(o_t^2, a_t^2)}$$

InstructGPT

Christiano et al. (2017)

Turned RLHF into the standard LLM recipe: (1) supervised fine-tuning on human demonstrations, (2) a reward model trained on human rankings of outputs, (3) PPO against that reward with a KL penalty to the SFT model, plus a pretraining-gradient mix ("PPO-ptx"). Its 1.3B model was preferred over the 175B GPT-3. The KL-regularized objective is DPO's starting point:

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{RL}}}\left[r_\theta(x,y) - \beta \log \frac{\pi_\phi^{RL}(y|x)}{\pi^{SFT}(y|x)}\right] + \gamma E_{x\sim D_{pretrain}}\left[\log \pi_\phi^{RL}(x)\right]$$

Direct Preference Optimization

Rafailov et al. (2023)

Fundamentally shifted LLM alignment by mathematically proving that the reward modeling phase and the RL optimization phase (like PPO) can be combined. It allows optimizing a policy directly from human preference data using a simple classification loss, completely eliminating the need for a separate reward model or an unstable actor-critic loop.

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

DeepSeekMath (GRPO)

Shao et al. (2024)

Introduced Group Relative Policy Optimization (GRPO). It builds on PPO but eliminates the need for a memory-intensive value model (critic) by normalizing rewards across a group of sampled outputs from the same prompt, heavily optimizing RL for mathematical reasoning in LLMs.

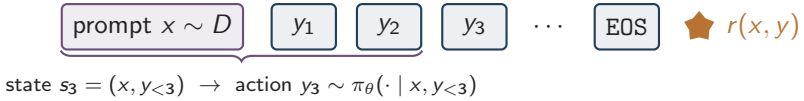
REINFORCE Leave-One-Out

Ahmadian et al. (2024)

The direct architectural precursor to DeepSeekMath's GRPO. It demonstrated that for LLMs, you can completely drop PPO's memory-heavy value critic and construct an unbiased baseline using the mean reward of other samples generated from the same prompt.

$$\hat{A}_i = R_i - \frac{1}{K-1}\sum_{j\neq i} R_j$$

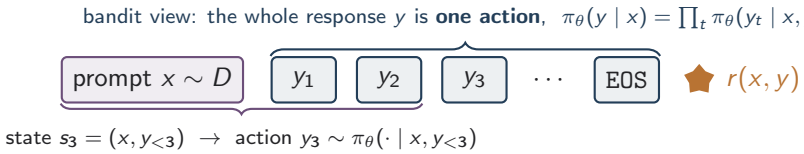
Text Generation as an RL Problem



Start distribution D
the prompts users send
— "D comes from the application"

Dynamics
append the token:
known and
deterministic

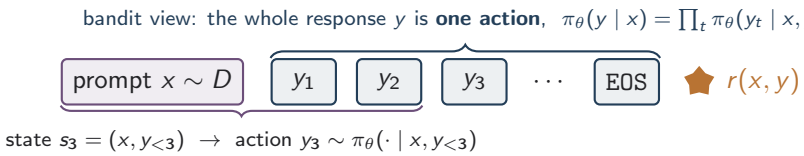
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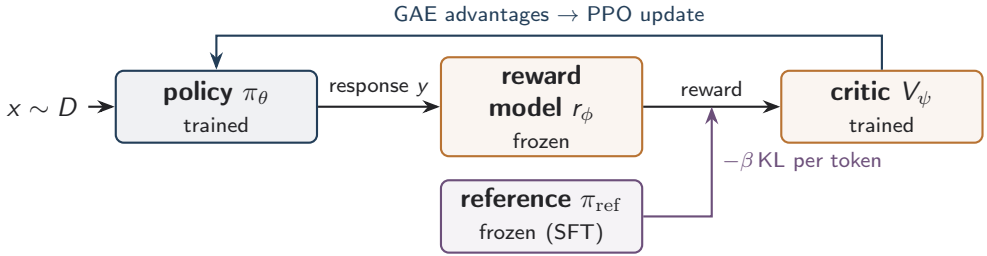


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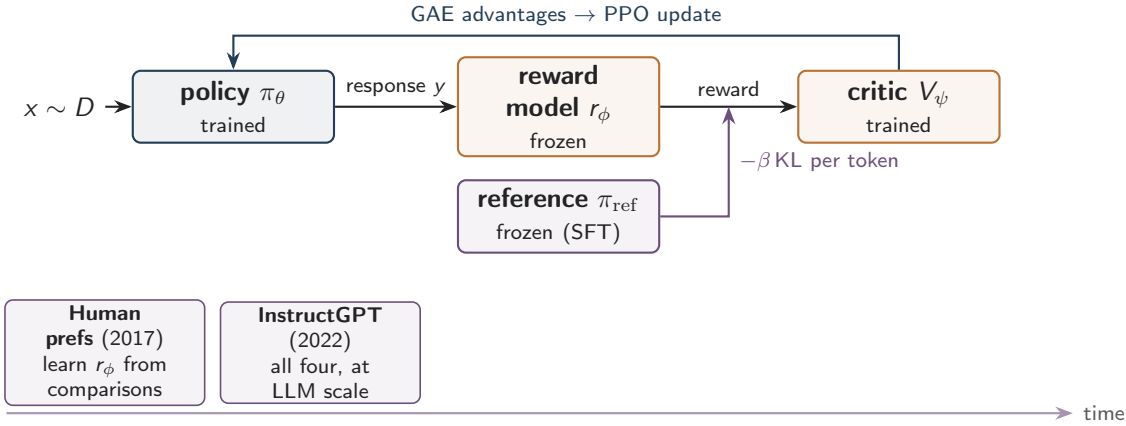
Dynamics
append the token: **known** and **deterministic**

Reward
only at the end — but who writes $r(x, y)$ for "helpful"?

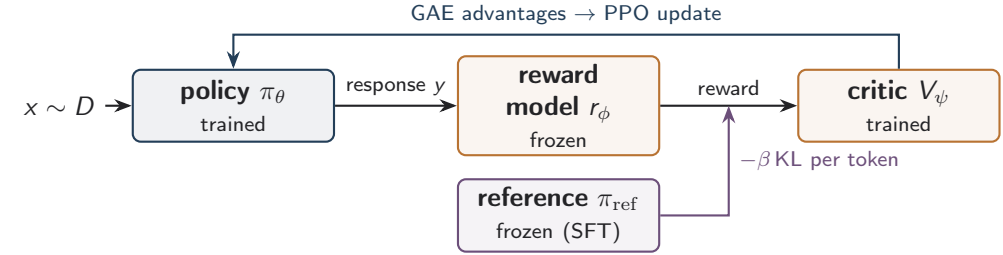
RLHF With PPO: What Can We Remove?



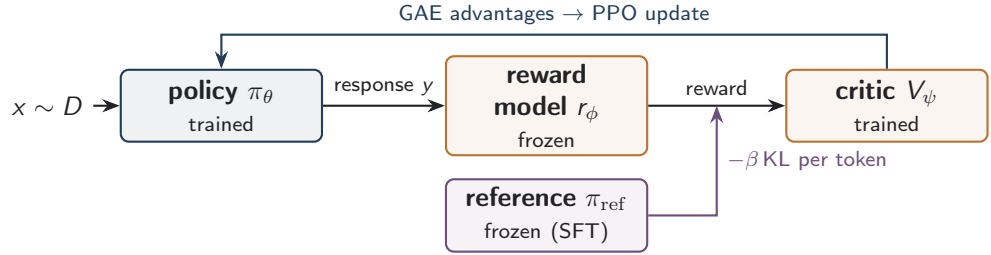
RLHF With PPO: What Can We Remove?



RLHF With PPO: What Can We Remove?



RLHF With PPO: What Can We Remove?



Deep Reinforcement Learning from Human Preferences

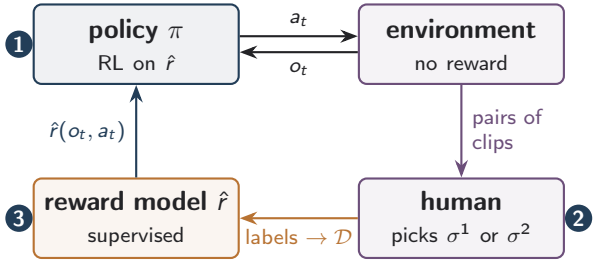
Christiano, Leike, Brown, Martic, Legg & Amodei
NeurIPS 2017

Introduces a **reward model** learned from pairwise human comparisons

Setting & Assumptions

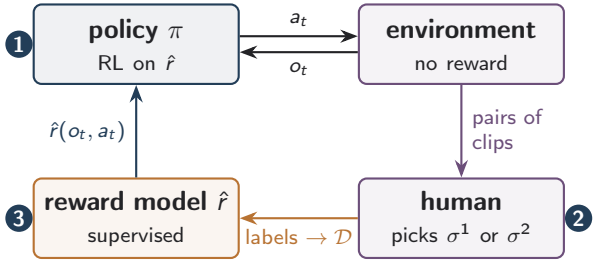
- Observations o_t, actions a_t (Atari, MuJoCo) — **no reward** from the environment.
- A human compares two short clips σ^1, σ^2 (1–2 s): which is better?
- Goal: behavior the human prefers, with **few** queries.
- No resets to chosen states: clips start wherever the agent was.
- The policy is trained by any RL method: **TRPO** (MuJoCo), **A2C** (Atari).

Three Processes, Running Asynchronously



1. RL maximizes the **predicted** reward $\sum_t \hat{r}(o_t, a_t)$.
2. Pairs of clips go to the human.
3. $\hat{r}$ is fit to all comparisons so far.

Three Processes, Running Asynchronously



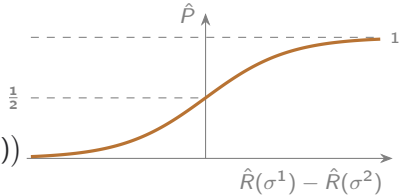
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2. Pairs of clips go to the human.
3. $\hat{r}$ is fit to all comparisons so far.

Feedback on $< 1\%$ of interactions. MuJoCo: 700 labels $\approx$ RL with the true reward. Backflip: 900 queries, < 1 hour.

Fitting the Reward: the Bradley–Terry Model

Preference = logistic in the **difference** of summed rewards:

$$\hat{P}[\sigma^1 \succ \sigma^2] = \frac{e^{\sum_t \hat{r}(o_t^1, a_t^1)}}{e^{\sum_t \hat{r}(o_t^1, a_t^1)} + e^{\sum_t \hat{r}(o_t^2, a_t^2)}} = \sigma\big(\hat{R}(\sigma^1) - \hat{R}(\sigma^2)\big)$$

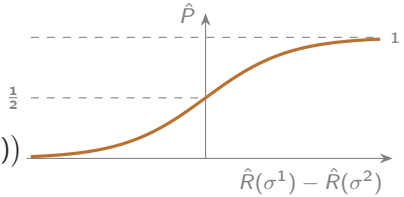


Like **Elo** in chess: the rating gap predicts who wins.

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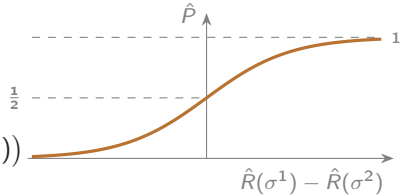
Fit $\hat{r}$ by cross-entropy on the labels μ ($\hat{P}_{12} = \hat{P}[\sigma^1 \succ \sigma^2]$):

$$\text{loss}(\hat{r}) = -\sum_{\mathcal{D}} \Big[\mu(1) \log \hat{P}_{12} + \mu(2) \log \hat{P}_{21}\Big]$$

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$$\text{loss}(\hat{r}) = -\sum_{\mathcal{D}} \Big[\mu(1) \log \hat{P}_{12} + \mu(2) \log \hat{P}_{21}\Big]$$

Only differences matter: $\hat{r} + c$ fits equally well $\Rightarrow$ normalize. This invariance is what DPO exploits.

Training Language Models to Follow Instructions with Human Feedback

Ouyang, Wu, Jiang, et al. (OpenAI)
NeurIPS 2022

Introduces **InstructGPT**: the SFT → reward model → PPO recipe

Setting & Assumptions

- D: real prompts sent to the OpenAI API, plus labeler-written ones.
- **Bandit** environment: a prompt, a response, a reward from the RM, end of episode.
- Rewards from a 6B **reward model** trained on labeler rankings.
- Per-token **KL penalty** to the SFT model.
- GPT-3 architecture: 1.3B, 6B, 175B.

The Three-Step Recipe

1. SFT

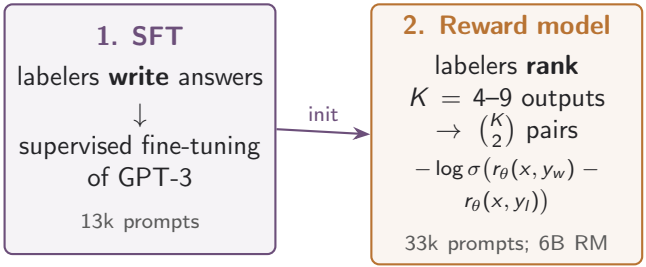
labelers **write** answers

↓

supervised fine-tuning of GPT-3

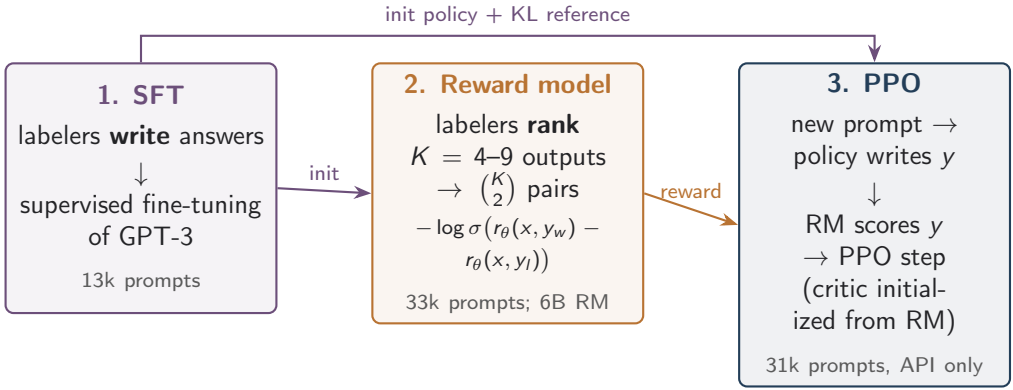
13k prompts

The Three-Step Recipe



Ranking K outputs at once: all $\binom{K}{2}$ pairs in **one batch element** — otherwise the RM overfits.

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The KL-Regularized Objective (PPO-ptx)

$$\text{objective}(\phi) = \mathbb{E}_{(x,y)\sim D_{\pi_\phi^{RL}}}\Big[\underbrace{r_\theta(x,y)}_{\text{reward model}} - \underbrace{\beta \log \frac{\pi_\phi^{RL}(y \mid x)}{\pi^{SFT}(y \mid x)}}_{\text{stay near SFT}}\Big] + \underbrace{\gamma\, \mathbb{E}_{x\sim D_{\text{pre}}}\big[\log \pi_\phi^{RL}(x)\big]}_{\text{pretraining mix}}$$

Reward: the RM is only a proxy for the labelers.

KL: per token, against **over-optimizing** the RM.

ptx: fixes the "alignment tax" on public NLP benchmarks.

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Labelers prefer the **1.3B** InstructGPT to the **175B** GPT-3. $85 \pm 3\%$ of the time.

175B vs. 175B: preferred

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Labelers prefer the **1.3B** InstructGPT to the **175B** GPT-3. 175B vs. 175B: preferred $85 \pm 3\%$ of the time.

Without ptx ($\gamma = 0$) this is the RLHF objective that **DPO** starts from. (γ here is not a discount.)

Direct Preference Optimization: Your Language Model Is Secretly a Reward Model

Rafailov, Sharma, Mitchell, Ermon, Manning & Finn
NeurIPS 2023

Introduces **DPO**: the RLHF objective solved with a **classification loss**

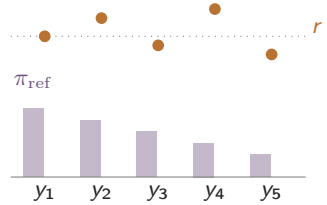
Setting & Assumptions

- Same objective as RLHF: reward minus $\beta\, D_{\mathrm{KL}}(\pi \,\|\, \pi_{\mathrm{ref}})$, with $\pi_{\mathrm{ref}} = \pi^{SFT}$.
- Preferences follow **Bradley–Terry** with a latent reward r^*.
- A **fixed** dataset $\mathcal{D} = \{(x, y_w, y_l)\}$ (offline).
- No reward model, **no sampling** from π_θ during training, no critic.

From the RLHF Objective to a Loss on the Policy

RLHF objective, for a reward r

$$\max_{\pi} \; \mathbb{E}_{x,\, y\sim\pi}\big[r(x,y)\big] - \beta\, D_{\mathrm{KL}}\big[\pi \,\|\, \pi_{\mathrm{ref}}\big]$$



Maximize reward, but pay β for every bit of KL away from π_{ref}.

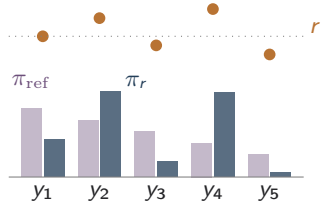
From the RLHF Objective to a Loss on the Policy

RLHF objective, for a reward r

$$\max_{\pi} \; \mathbb{E}_{x,\,y\sim\pi}\big[r(x,y)\big] - \beta\, D_{\mathrm{KL}}\big[\pi \,\|\, \pi_{\mathrm{ref}}\big]$$

its optimum in closed form: tilt π_{ref}

$$\pi_r(y|x) = \frac{1}{Z(x)}\,\pi_{\mathrm{ref}}(y|x)\,e^{r(x,y)/\beta}$$



Reweight π_{ref} by $e^{r/\beta}$, normalize by $Z(x) = \sum_y \pi_{\mathrm{ref}} e^{r/\beta}$ — intractable.

From the RLHF Objective to a Loss on the Policy

RLHF objective, for a reward r

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$$\pi_r(y|x) = \frac{1}{Z(x)}\, \pi_{\mathrm{ref}}(y|x)\, e^{r(x,y)/\beta}$$

solve for the reward

$$r(x,y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\mathrm{ref}}(y|x)} + \beta \log Z(x)$$

Read it backwards: **every policy defines a reward**

$$\hat{r}_\theta(x,y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\mathrm{ref}}(y|x)}$$

up to a term $\beta \log Z(x)$ that depends on x only.

From the RLHF Objective to a Loss on the Policy

RLHF objective, for a reward r

$$\max_{\pi} \; \mathbb{E}_{x,\,y\sim\pi}\big[r(x,y)\big] - \beta\, D_{\mathrm{KL}}\big[\pi \,\|\, \pi_{\mathrm{ref}}\big]$$

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$$\pi_r(y|x) = \frac{1}{Z(x)}\, \pi_{\mathrm{ref}}(y|x)\, e^{r(x,y)/\beta}$$

solve for the reward

$$r(x,y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\mathrm{ref}}(y|x)} + \beta \log Z(x)$$

into Bradley–Terry: $Z(x)$ cancels

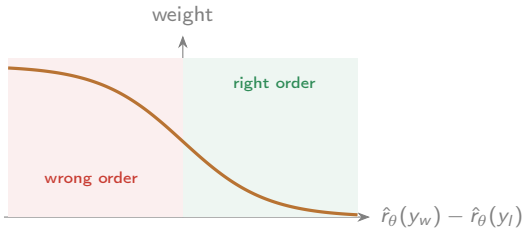
$$\mathcal{L}_{\mathrm{DPO}} = -\mathbb{E}_{\mathcal{D}}\Big[\log \sigma\big(\hat{r}_\theta(x,y_w) - \hat{r}_\theta(x,y_l)\big)\Big]$$

Bradley–Terry compares two responses to the **same** $x \Rightarrow$ $\beta \log Z(x)$ drops out, with $\hat{r}_\theta = \beta \log \frac{\pi_\theta}{\pi_{\mathrm{ref}}}$.

The reward-model loss, with r_ϕ replaced by $\hat{r}_\theta$:
the LM is the reward model.

What the DPO Update Does

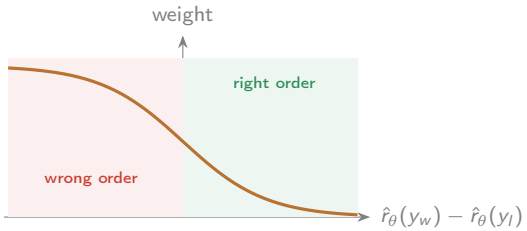
$$\nabla_\theta \mathcal{L}_{\mathrm{DPO}} = -\beta\, \mathbb{E}_{\mathcal{D}}\Big[\underbrace{\sigma\big(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)\big)}_{\text{large when the ranking is wrong}} \Big(\underbrace{\nabla_\theta \log \pi_\theta(y_w|x)}_{\text{raise } y_w} - \underbrace{\nabla_\theta \log \pi_\theta(y_l|x)}_{\text{lower } y_l} \Big) \Big]$$



- Like the reward-model loss — on the policy itself.
- Weight fades once a pair is ranked right. Without it (unlikelihood) the model **degenerates**.

What the DPO Update Does

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- Like the reward-model loss — on the policy itself.
- Weight fades once a pair is ranked right. Without it (unlikelihood) the model **degenerates**.
- Sentiment task: best reward for every KL budget, above PPO.
- TL;DR summaries: 61% win rate vs. PPO's 57%.

DPO: What We Removed, and What It Costs

Removed

- The reward model — implicit in π_θ.
- The critic and the RL loop.
- Sampling during training.

Left: π_θ, π_{ref} and one supervised-style loss. Same optimum as RLHF (under Bradley–Terry).

Costs

- **Offline**: the responses come from π^{SFT}, not from π_θ — no exploration.
- Learns only on the pairs in $\mathcal{D}$.
- Later evidence: online beats offline (DeepSeekMath); DPO answers are long (RLOO paper).

Part I echo: CPI's quality depends on how well the sampling distribution covers the good states ($\|d_{\pi^*}/\mu\|_\infty$). Here: how well π^{SFT}'s answers cover good answers.

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

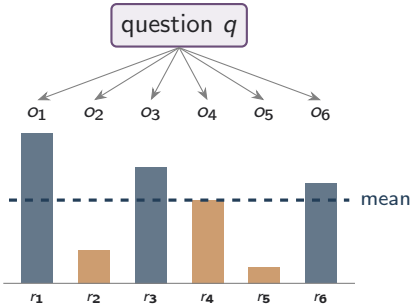
Shao, Wang, Zhu, et al.
(DeepSeek-AI)
arXiv 2024

Introduces **GRPO**: PPO without the critic

Setting & Assumptions

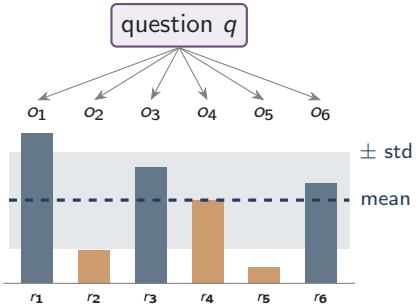
- Math questions q (GSM8K, MATH; chain of thought), policy: DeepSeekMath-Instruct 7B.
- A **reward model** scores each output o — at the end (outcome) or per step (process).
- Online: $G = 64$ outputs per question from the current policy.
- Problem with PPO's critic: **as large as the policy**, and hard to fit per token when the reward comes only at the end.

Group-Relative Advantages



Sample G outputs for the **same** question; score them.

Group-Relative Advantages

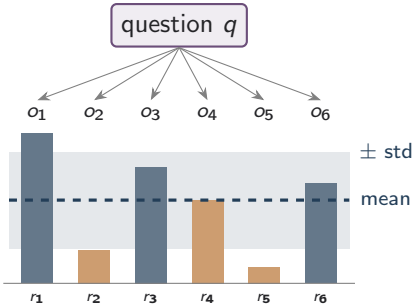


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$$\hat{A}_{i,t} = \frac{r_i - \mathrm{mean}(\mathbf{r})}{\mathrm{std}(\mathbf{r})}$$

the same value for every token t of o_i

Group-Relative Advantages



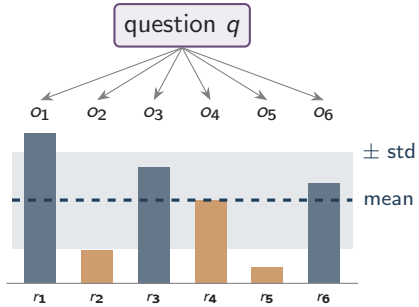
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- The group mean replaces $V(q)$: a critic computed **on the fly**.
- Matches how RMs are trained: **comparisons** within one question.

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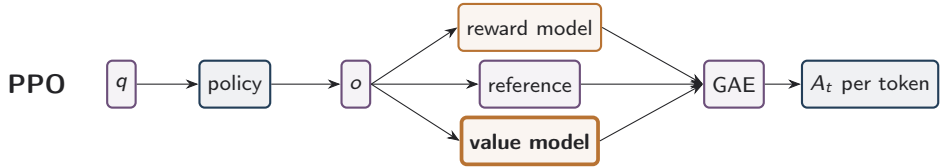
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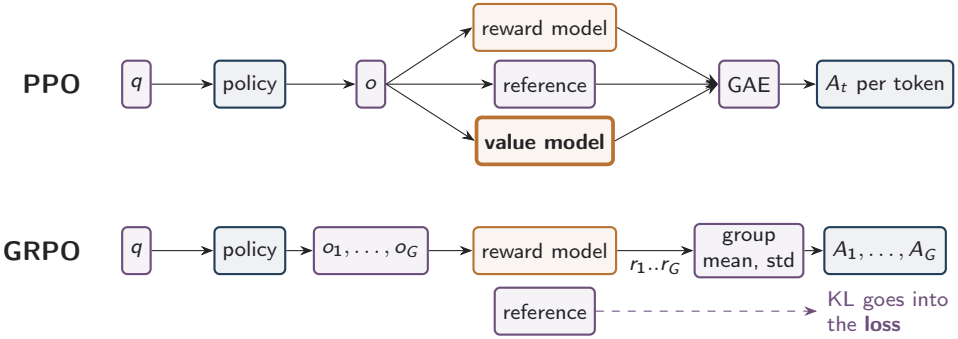
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RL on GSM8K + MATH questions only: GSM8K 82.9 → 88.2%, MATH 46.8 → 51.7%, and gains out of domain.

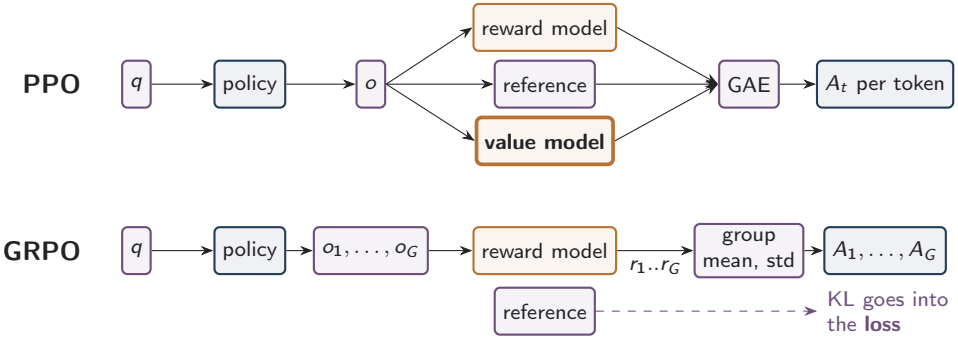
PPO vs. GRPO


PPO
q
policy
o
reward model
reference
value model
GAE
A_t per token

PPO vs. GRPO


PPO
q
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A_t per token
GRPO
q
policy
$o_1, \ldots, o_G$
reward model
$r_1..r_G$
group mean, std
$A_1, \ldots, A_G$
reference
KL goes into the loss

PPO vs. GRPO



$$\mathcal{J}_{\mathrm{GRPO}}(\theta) = \mathbb{E}\Big[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big\{\min\left(\rho_{i,t}\hat{A}_{i,t},\ \mathrm{clip}(\rho_{i,t}, 1\pm\varepsilon)\,\hat{A}_{i,t}\right) - \beta\, D_{\mathrm{KL}}\left[\pi_\theta \| \pi_{\mathrm{ref}}\right]\Big\}\Big]$$

$\rho_{i,t} = \pi_\theta(o_{i,t}|q, o_{i,<t}) / \pi_{\theta_{\mathrm{old}}}(o_{i,t}|q, o_{i,<t})$: still PPO's clipped ratio, per token.

Back to Basics: Revisiting REINFORCE-Style Optimization for Learning from Human Feedback in LLMs

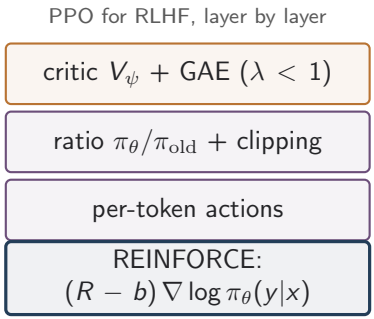
Ahmadian, Cremer, Gallé, et al. (Cohere)
ACL 2024

Shows that **REINFORCE** with a leave-one-out baseline beats PPO for RLHF

Setting & Assumptions

- Standard RLHF: SFT $\rightarrow$ reward model r_ϕ $\rightarrow$ online RL.
- KL-shaped reward per response: $R(x, y) = r_\phi(x, y) - \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$.
- **Claim**: PPO's tools fight high variance from a **weak** start. A pretrained LLM is a **strong** start.
- Reward only at `EOS`, deterministic dynamics $\Rightarrow$ treat y as **one action**.

Which Parts of PPO Does RLHF Need?



Which Parts of PPO Does RLHF Need?

PPO for RLHF, layer by layer

~~critic V_ψ + GAE ($\lambda < 1$)~~

ratio $\pi_\theta / \pi_{\text{old}}$ + clipping

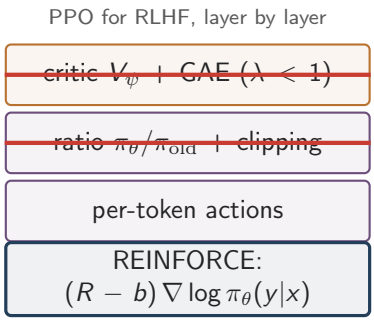
per-token actions

REINFORCE:
$(R - b) \nabla \log \pi_\theta(y|x)$

Bias for variance? Not worth it.

Reward falls steadily as λ goes $1 \to 0$; $\lambda = 1$ (no bootstrapping) is best.

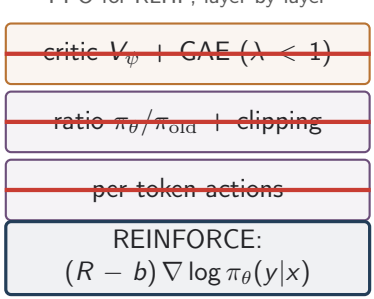
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Clipped in $< 5\%$ of each batch; removing ratio and clip does not hurt.

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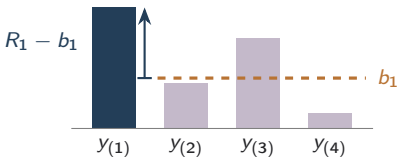


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Tokens as actions? Not needed.
The whole response as one action works as well or better.

The Leave-One-Out Baseline



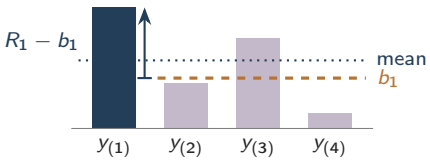
b_1 = mean of the **other** samples

k samples $y_{(i)} \sim \pi_\theta(\cdot|x)$, rewards $R_i = R(y_{(i)}, x)$:

$$\frac{1}{k}\sum_{i=1}^{k}\Big[R_i - \underbrace{\frac{1}{k-1}\sum_{j\neq i} R_j}_{b_i}\Big]\nabla \log \pi_\theta(y_{(i)}|x)$$

Baseline independent of $y_{(i)} \Rightarrow$ **unbiased**.

The Leave-One-Out Baseline



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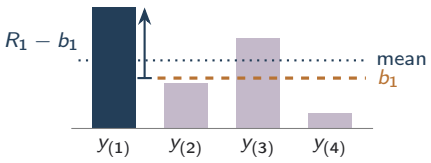
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RLOO ($k = 4$) beats PPO's win rate by 10–32 points. Three models, no critic.

What Did We Remove?

	Reward model	Critic	KL to π_{ref}	Samples of π_θ	Clip
InstructGPT (PPO)	✓	✓	✓ in reward	✓	✓

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RLOO	✓	× leave-one-out	✓ in reward	✓ k per x	×

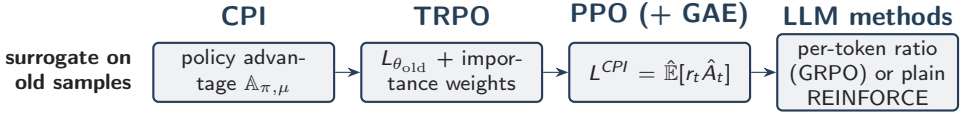
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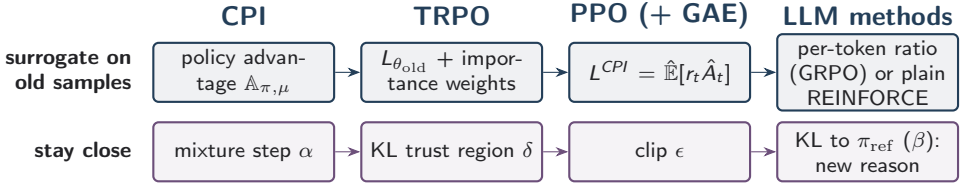
Never removed: the KL to π_{ref} — the learned reward is only trusted near π^{SFT}.

Why the critic could go: many samples per prompt are cheap, and the prompt is the only real state.

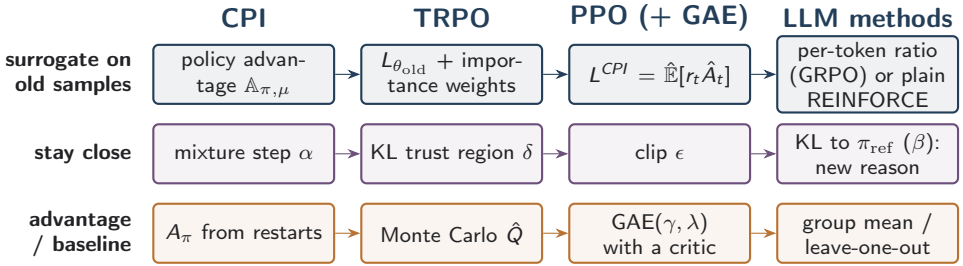
The Thread: From CPI to LLMs



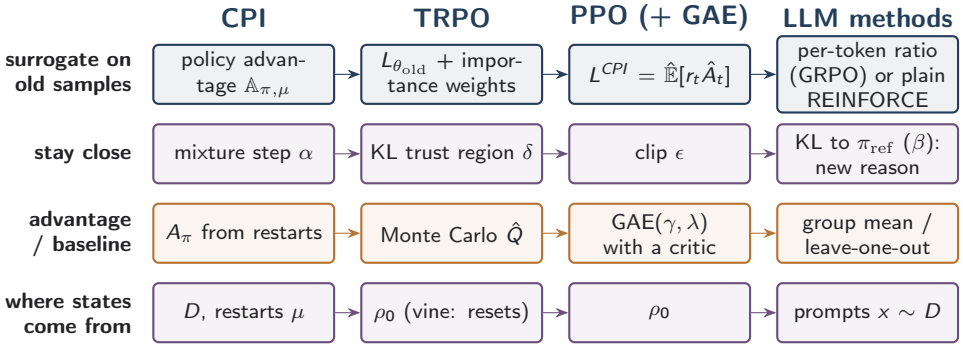
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The Thread: From CPI to LLMs



Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (1/3)

The conservative mixtures form a **one-parameter policy class** $\{\pi_\alpha\}_{\alpha \in [0,1]}$:

$$\pi_\alpha(a;s) = (1-\alpha)\,\pi(a;s) + \alpha\,\pi'(a;s) \quad \Longrightarrow \quad \frac{\partial \pi_\alpha(a;s)}{\partial \alpha} = \pi'(a;s) - \pi(a;s)$$

So the policy gradient formula applies as is — α **plays the role of** θ:

Policy gradient, any $\theta \in \mathbb{R}^m$: $\nabla_\theta \eta_\mu = \dfrac{1}{1-\gamma} \sum_s d_{\pi_\theta,\mu}(s) \sum_a \nabla_\theta \pi_\theta(a;s)\, Q_{\pi_\theta}(s,a)$

$\theta := \alpha$

Mixture, at $\alpha = 0$ $(\pi_0 = \pi)$: $\left.\dfrac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \dfrac{1}{1-\gamma} \sum_s d_{\pi,\mu}(s) \sum_a \dfrac{\partial \pi_\alpha(a;s)}{\partial \alpha}\, Q_\pi(s,a)$

The factor $\frac{1}{1-\gamma}$ comes from our normalization: $d_{\pi,\mu}$ sums to 1 and Q_π carries a $(1-\gamma)$.

Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (2/3)

Substitute the derivative of the mixture:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

Since $\sum_a \pi(a;s)\, Q_\pi(s,a) = V_\pi(s)$, $\sum_a \pi'(a;s) = 1$ and $A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$, the inner sum is:

$$\sum_a \pi'(a;s)\, Q_\pi(s,a) - V_\pi(s) = \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Back into the expectation — this is the policy advantage $\mathbb{A}_{\pi,\mu}(\pi')$:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] = \frac{1}{1-\gamma}\,\mathbb{A}_{\pi,\mu}(\pi')$$

Proof: α and θ Play the Same Role (3/3)

CPI: parameter $\alpha \in [0, 1]$

$$\eta_\mu(\pi_\alpha) = \eta_\mu(\pi) + \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

old-state surrogate exact to 1st order (this proof)

step size: α **itself**

generalize

TRPO: parameter $\theta \in \mathbb{R}^m$

$$\eta(\pi_\theta) = L_{\pi_{\theta_0}}(\pi_\theta) + o(\|\theta - \theta_0\|)$$

old-state surrogate exact to 1st order (TRPO Eq. 4)

step size (Thm. 1): $\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}})$

The proof never used that α is a mixing weight — only that it parameterizes the policy. TRPO keeps the first-order argument for any θ, and replaces "how far" (α) by a distance between policies.

◂ Back to the policy advantage

Comparing With Bertsekas & Tsitsiklis (1/5): A Worst Case at Every State

$\bar{A}$ as in Lemma 6.1, $\Delta\eta = \eta_\mu(\pi') - \eta_\mu(\pi)$. Given: $\varepsilon = \max_s |\tilde{V}(s) - V_\pi(s)|$ and π' greedy w.r.t. $\tilde{V}$, i.e. $T_{\pi'}\tilde{V} \geq T_\pi \tilde{V}$,
where $(T_{\pi'} V)(s) = \mathbb{E}_{a\sim\pi'}\left[(1-\gamma)\mathcal{R}(s,a) + \gamma\, \mathbb{E}_{s'} V(s')\right]$, so $\bar{A} = T_{\pi'} V_\pi - V_\pi$.

$T_{\pi'} V_\pi$

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$$T_{\pi'}V_\pi \;\geq\; T_{\pi'}\tilde{V} - \gamma\varepsilon$$

V_π vs. $\tilde{V}$ at the next state: error $\leq \varepsilon$, times γ

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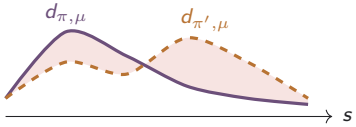
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$$

ε = **how wrong our values are**. It only limits how **negative** $\bar{A}$ can be; nothing positive is known, since π' is greedy for the *wrong* values. ($\varepsilon = 0$: $\bar{A} \geq 0$ everywhere — exact policy improvement.)

Comparing With Bertsekas & Tsitsiklis (2/5): Thm 4.1 at $\alpha = 1$

Split the new states into old states + shift:

$$(1-\gamma)\Delta\eta = \sum_s d_{\pi',\mu}(s)\,\bar{A}(s)$$

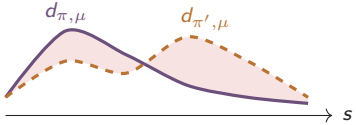


shaded: the shift — $\bar{A}$ unknown there.

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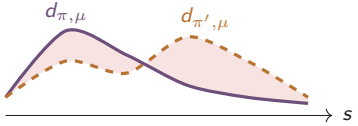


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Comparing With Bertsekas & Tsitsiklis (2/5): Thm 4.1 at $\alpha = 1$

Split the new states into old states + shift:

$$(1-\gamma)\Delta\eta = \sum_s d_{\pi',\mu}(s)\,\bar{A}(s) \;=\; \underbrace{\sum_s d_{\pi,\mu}(s)\,\bar{A}(s)}_{=\mathbb{A}\ \text{(measurable)}} + \underbrace{\sum_s \big(d_{\pi',\mu} - d_{\pi,\mu}\big)(s)\,\bar{A}(s)}_{\text{shift}}$$



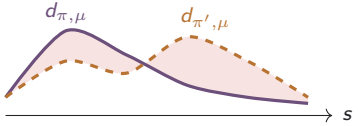
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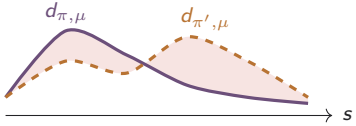
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$$\Delta\eta \;\geq\; \frac{\mathbb{A}}{1-\gamma} - \frac{2\gamma\varepsilon}{1-\gamma} \qquad = \text{Thm 4.1 at } \alpha = 1$$

Comparing With Bertsekas & Tsitsiklis (3/5): Why the Mixture Helps

Same split for $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi'$:

- Per state: $\bar{A}_{\text{new}}(s) = \alpha\,\bar{A}(s)$
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- Shift: π_{new} acts like π w.p. $1-\alpha$ each step,
  Pr(deviated by t) $\leq 1-(1-\alpha)^t$

$$\Rightarrow \quad \|d_{\pi_{\text{new}},\mu} - d_{\pi,\mu}\|_1 \leq \frac{2\alpha\gamma}{1-\gamma(1-\alpha)}$$

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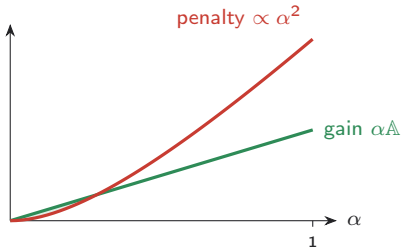
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Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$.

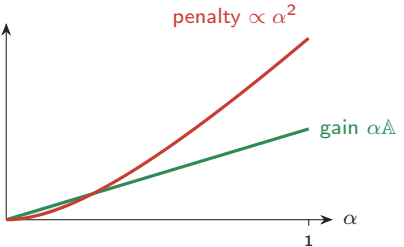
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- Multiply: $(1-\gamma)\Delta\eta \ge \alpha\mathbb{A} - \alpha\varepsilon \cdot \dfrac{2\alpha\gamma}{1-\gamma(1-\alpha)}$ = Thm 4.1



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$.

Small α: the gain wins. $\alpha = 1$: since $\mathbb{A} \le \varepsilon$, we need $\mathbb{A} > 2\gamma\varepsilon$ — impossible for $\gamma \ge \frac{1}{2}$. The positive term alone does not justify the full greedy step; **the small step does**.

Comparing With Bertsekas & Tsitsiklis (4/5): Same Penalty, Different ε

	B&T	K&L, Thm 4.1
$\varepsilon =$	$\max_s \lvert\tilde{V}(s) - V_\pi(s)\rvert$	$\max_s \lvert\mathbb{E}_{a\sim\pi'} A_\pi(s,a)\rvert$
Depends on	our estimate $\tilde{V}$	the pair π, π'
Measures	how wrong our values are	largest one-step gain/loss of π'
Step size	fixed: full greedy step	α (mixture weight)

Comparing With Bertsekas & Tsitsiklis (4/5): Same Penalty, Different ε

	B&T	K&L, Thm 4.1	TRPO, Thm 1
$\varepsilon =$	$\max_s \lvert\tilde{V}(s) - V_\pi(s)\rvert$	$\max_s \lvert\mathbb{E}_{a\sim\pi'} A_\pi(s,a)\rvert$	$\max_{s,a} \lvert A_\pi(s,a)\rvert$
Depends on	our estimate $\tilde{V}$	the pair π, π'	π only
Measures	how wrong our values are	largest one-step gain/loss of π'	scale of the advantages
Step size	fixed: full greedy step	α (mixture weight)	$\alpha = D_{\mathrm{TV}}^{\max}(\pi, \tilde{\pi})$

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- **Common:** the price of not knowing $\bar{A}$ on the states the new policy visits.

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- **Common:** the price of not knowing $\bar{A}$ on the states the new policy visits.
- **Why it changed:** in B&T, π' is fixed by $\tilde{V}$, so only the estimate can go wrong. K&L ask a different question — *how far may we step?* — so ε measures the step, not the estimate.

Comparing With Bertsekas & Tsitsiklis (5/5): Where the Error Went, What TRPO Does

Where did B&T's approximation error go? Not into Thm 4.1's ε, but into

- the estimate of $\mathbb{A}$ from μ-restarts (sampling error, Hoeffding);
- the accuracy of the greedy chooser G_ε (CPI slide).

Both are **averages** under $d_{\pi,\mu}$ — much weaker than B&T's $\max_s$.

Back to Thm 4.1

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TRPO splits K&L's ε into a step and a scale:

$$|\bar{A}(s)| \;\le\; 2 \underbrace{D_{\mathrm{TV}}\big(\pi(\cdot|s), \tilde{\pi}(\cdot|s)\big)}_{\text{policy change} \le \alpha} \cdot \underbrace{\max_{s,a} |A_\pi(s,a)|}_{\text{advantage scale}}$$

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The step becomes a **distance between policies** that the algorithm controls ⇒ trust region.

Back to Thm 4.1

From $\mathbb{E}_{s\sim d}$ to a Sum Over Time

$$\frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{\pi',\mu}}[\bar{A}(s)] = \frac{1}{1-\gamma}\sum_{s} d_{\pi',\mu}(s)\,\bar{A}(s)$$

Back to the lemma

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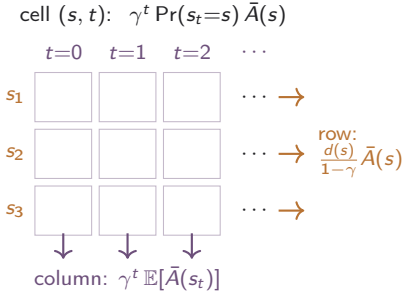
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Sum the grid by rows (step 2)

or by columns (step 3): same total.

◂ Back to the lemma

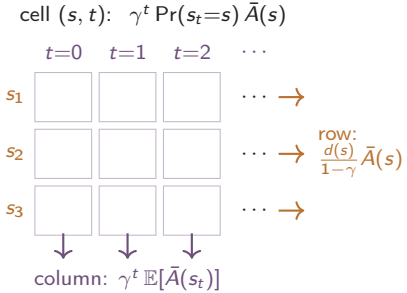
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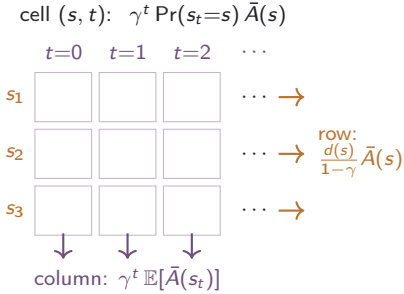
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$\mathbb{E}[\bar{A}(s_t)]$: over $s_t \sim p_t$, $p_t(s) = \Pr(s_t{=}s \mid s_0 \sim \mu, \pi')$ — the state at time t, **not discounted**.
$\mathbb{E}[\sum_t \cdots]$: over trajectories $s_0 \sim \mu$, $a_t \sim \pi'(\cdot|s_t)$, $s_{t+1} \sim P(\cdot|s_t, a_t)$.
The discount sits only in the weights γ^t.

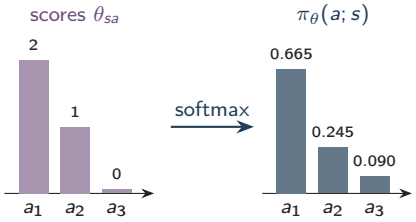


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◂ Back to the lemma

The Gibbs (Softmax) Policy



One state s, three actions. Every action keeps **some** probability.

One parameter per state–action pair ("table lookup"):

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}$$

- **Why "Gibbs":** the Boltzmann distribution of physics, $p \propto e^{-E/T}$, with $\theta = -E/T$.
- **Always stochastic:** $\pi_\theta(a;s) > 0$; deterministic only as $\theta \to \pm\infty$.
- **Rich:** any policy with full support, in every state independently.
- **Simple score:** $\partial_{\theta_{sb}} \log \pi_\theta(a;s) = 1[a=b] - \pi_\theta(b;s)$

▸ Derivation

◂ Back to policy gradients

Deriving the Gibbs Score

$$\pi(a) = \frac{e^{\theta_a}}{\sum_c e^{\theta_c}}$$

Back to the Gibbs policy

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Back to the Gibbs policy

Deriving the Gibbs Score

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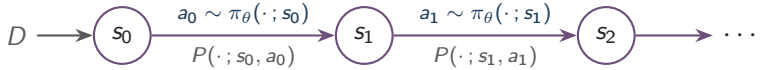
$$\log \pi(a) = \theta_a - \log \sum_c e^{\theta_c}$$

$$\partial_{\theta_b} \log \pi(a) = \underbrace{\partial_{\theta_b} \theta_a}_{1[a=b]} - \underbrace{\frac{e^{\theta_b}}{\sum_c e^{\theta_c}}}_{\pi(b)}$$

Back to the Gibbs policy

Why the Log-Derivative Trick Is Unbiased

θ sits in the **sampling distribution**, not in $R(\tau)$, so averaging $\nabla_\theta R$ gives nothing.



$$p_\theta(\tau) = D(s_0) \prod_t \pi_\theta(a_t; s_t)\, P(s_{t+1}; s_t, a_t)$$

$$\nabla_\theta \eta_D = \sum_\tau \underbrace{\nabla_\theta\, p_\theta(\tau)}_{p_\theta(\tau)\,\nabla_\theta \log p_\theta(\tau)} \qquad R(\tau) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\,\nabla_\theta \log p_\theta(\tau)\big]$$

$$\nabla_\theta \log p_\theta(\tau) = \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$$

(log turns $\prod$ into $\sum$; grey terms have no θ)

$\hat{g} = R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$, $\tau \sim \pi_\theta \;\Rightarrow\; \mathbb{E}[\hat{g}] = \nabla_\theta \eta_D$ **exactly**, and no need to know P.

Back to PG = SGA

CPI Through the SGD Lens

Take the table $\pi(a; s)$ itself as the parameter: CPI is a **first-order method** on η_μ.

	SGD (non-convex, smooth)	CPI
direction	$\hat{g} \approx \nabla J(w)$	greedy $\pi' = G_\varepsilon(\pi, \mu)$: best linearized direction
step	$w \leftarrow w + a_k\, \hat{g}$	$\pi \leftarrow (1-\alpha)\,\pi + \alpha\,\pi'$ (Frank–Wolfe step)
step size	$\sum_k a_k = \infty,\ \sum_k a_k^2 < \infty$	$\alpha \propto \hat{\mathbb{A}}$: slope over a smoothness bound
noise	averages out over many steps	each step accurate w.h.p. (Hoeffding)
stationary	$\|\nabla J(w_R)\| \le \epsilon$, random iterate	$\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$, last iterate
cost	$O(\sigma^2/\epsilon^4)$ samples	$72R^2/\varepsilon^2$ updates $\times$ $\tilde{O}(R^2/\varepsilon^2)$ rollouts
global?	$\times$ a stationary point can be bad	✓ Thm. 6.2, up to the mismatch factor

Back to the three questions

CPI Through the SGD Lens

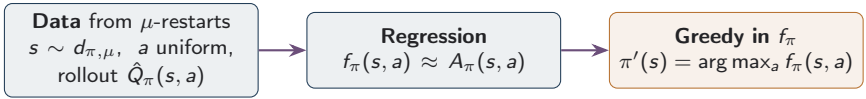
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Same skeleton: a bounded objective and a sure gain per step ⇒ few steps to a stationary point. CPI adds Thm. 6.2: **stationary ⇒ near-optimal**.

Back to the three questions

Building G_ε by Regression on Advantages



Required accuracy (Sec. 7.1):

$$L(f_\pi) = \underbrace{\mathbb{E}_{s\sim d_{\pi,\mu}}}_{\text{average over states}} \underbrace{\max_a}_{\text{worst action}} \left|A_\pi(s,a) - f_\pi(s,a)\right| \le \frac{\varepsilon}{2}$$

Why it suffices: with $e(s) = \max_a |A_\pi - f_\pi|$, $\hat{a} = \arg\max_a f_\pi$, $a^* = \arg\max_a A_\pi$:

$$A_\pi(s,\hat{a}) \ge f_\pi(s,\hat{a}) - e(s) \ge f_\pi(s,a^*) - e(s) \ge A_\pi(s,a^*) - 2e(s)$$

$$\mathbb{E}_{s\sim d_{\pi,\mu}}: \quad \mathbb{A}_{\pi,\mu}(\pi') \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - 2L \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - \varepsilon$$

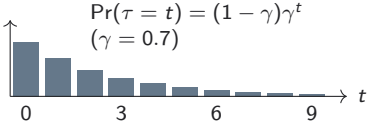
An **average** (ℓ_1) error over $d_{\pi,\mu}$, not the $\max_s$ (ℓ_∞) error of greedy DP (Eq. 3.1): rarely visited states may be fit badly.

Back to CPI

Stopping With Probability $1-\gamma$ Samples $d_{\pi,\mu}$

Run π from $s_0 \sim \mu$; after each step stop w.p. $1-\gamma$, independently of the trajectory. Return s_τ.

$\Pr(\tau = t) = \gamma^t(1-\gamma)$



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

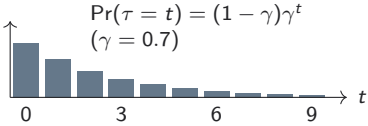
◂ Back to estimating 𝔸

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$$\Pr(\tau = t) = \gamma^t(1-\gamma)$$

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The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

◂ Back to estimating 𝔸

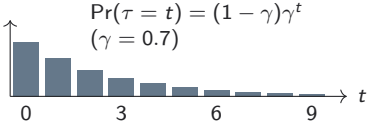
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◂ Back to estimating $\mathbb{A}$

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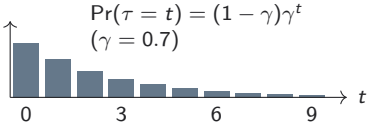
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Step 2 needs independence: τ does not look at the states.
The $(1-\gamma)$ in $d_{\pi,\mu}$ is what makes $\sum_t \Pr(\tau = t) = 1$.
γ = probability of surviving one more step;
$\mathbb{E}[\tau] = \gamma/(1-\gamma)$.



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

◂ Back to estimating $\mathbb{A}$

Hoeffding's Inequality and the Estimate of $\mathbb{A}$

If $X_1, \ldots, X_k$ are **independent** and $X_i \in [a, b]$ a.s., then for $\bar{X} = \frac{1}{k} \sum_i X_i$:

$$\Pr\left(|\bar{X} - \mathbb{E}\bar{X}| > \Delta\right) \;\leq\; 2\exp\Big(-\frac{2k\Delta^2}{(b-a)^2}\Big)$$

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Conditions for x_i:

- **Independent:** each x_i comes from a fresh μ-restart.
- **Bounded:** $\hat{Q}_i \in [0, R]$, $|\pi' - \pi| \leq 1 \Rightarrow x_i \in [-n_a R,\, n_a R]$, $b - a = 2n_a R$.
- **Right mean:** $\mathbb{E}x_i = \mathbb{A}$ (uniform a, weight n_a); bias of a finite rollout kept below $\varepsilon/6$.

◂ Back to estimating $\mathbb{A}$

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$$\Pr\left(|\hat{\mathbb{A}} - \mathbb{A}| > \Delta\right) \;\le\; 2\exp\Big(-\frac{2k\Delta^2}{4n_a^2R^2}\Big) = 2e^{-k\Delta^2/(2n_a^2R^2)}$$

$$k \;\ge\; \frac{2n_a^2R^2}{\Delta^2}\log\frac{2}{\delta} \quad \text{gives error } \le \Delta \text{ w.p. } \ge 1-\delta$$

Back to estimating $\mathbb{A}$

Tool: Importance Sampling

The estimate of $\mathbb{A}$ needs a **sum** over actions, but each rollout tries **one** action.

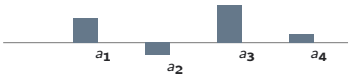
Sample from q, reweight by $w = p/q$

(needs $q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

$f(a)$

each a drawn w.p. $q = 1/n_a = \frac{1}{4}$



We can evaluate p pointwise, but only draw samples from q.

◂ Back to estimating $\mathbb{A}$

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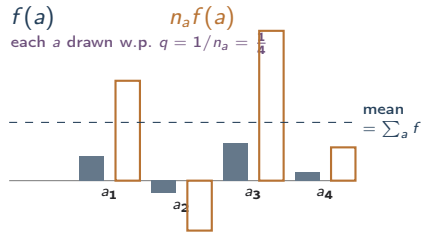
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A plain sum is the case $p \equiv 1$; CPI takes q uniform

$$\sum_a f(a) = \mathbb{E}_{a\sim q}\left[\frac{f(a)}{q(a)}\right] = \mathbb{E}_{a\sim \text{unif}}\big[n_a f(a)\big]$$



Average of the orange bars under uniform a = sum of the blue bars.

◂ Back to estimating $\mathbb{A}$

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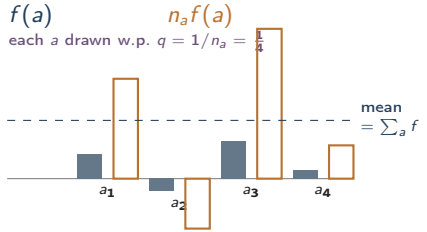
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Unbiased, but the weight stretches the range

$|f| \le R \;\Rightarrow\; |n_a f| \le n_a R \;\Rightarrow\; n_a^2$ in Hoeffding



One sampled bar can be n_a times taller ⇒ more samples needed. TRPO reuses this with $p = \pi_\theta$, $q = \pi_{\theta_{\text{old}}}$.

◂ Back to estimating $\mathbb{A}$

Why Uniform Actions? Choosing q

At a state $s \sim d_{\pi,\mu}$, the same target can be sampled two ways:

$$\sum_a (\pi' - \pi)(a; s)\, Q_\pi(s, a) \;=\; \underbrace{\mathbb{E}_{a\sim\text{unif}}\big[n_a(\pi' - \pi)(a; s)\, Q_\pi(s, a)\big]}_{\text{importance sampling, } p\equiv 1} \;=\; \underbrace{\mathbb{E}_{a\sim\pi'}\big[Q_\pi(s, a)\big] - V_\pi(s}_{\text{sample from } \pi'}$$

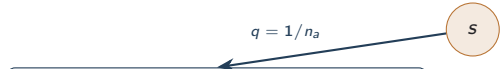


Back to estimating 𝔸

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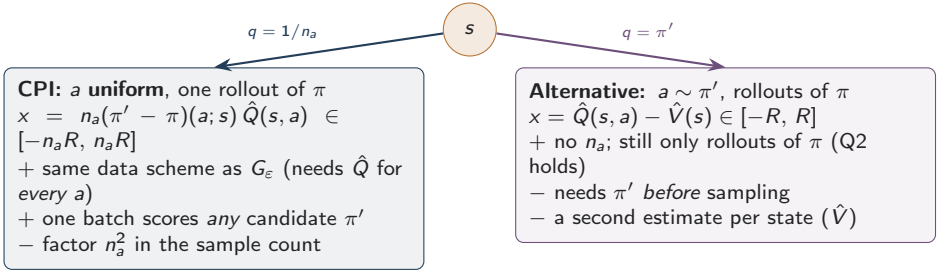
CPI: *a* **uniform**, one rollout of π
$x \;=\; n_a(\pi' - \pi)(a;s)\, \hat{Q}(s,a) \;\in$
$[-n_a R,\, n_a R]$
$+$ same data scheme as G_ε (needs $\hat{Q}$ for *every* a)
$+$ one batch scores *any* candidate π'
$-$ factor n_a^2 in the sample count

Back to estimating $\mathbb{A}$

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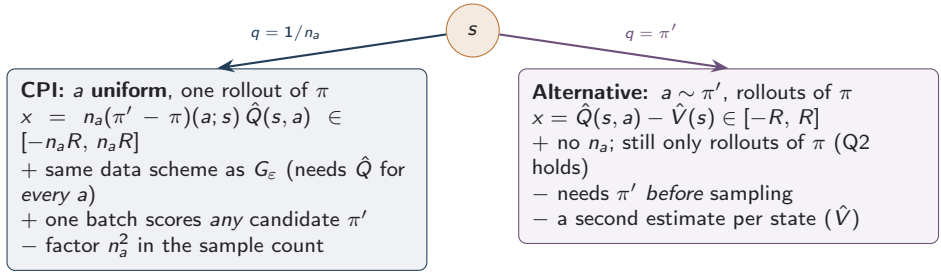


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Not plain IS with $q = \pi'$: weight $1/\pi'(a)$, but greedy π' is often deterministic, so $\pi'(a) = 0$ where $\pi(a)Q_\pi(s,a) \neq 0$ and the identity fails.

Back to estimating 𝔸